HyLike: Hyperspectral Anomaly Detection via Pixel-Wise Likelihood Modeling With Transferred Natural Image Priors

Qingkang He, Lianru Gao, *Senior Member, IEEE*, Michael K. Ng, *Senior Member, IEEE*, Yao Liu, and Lina Zhuang, *Member, IEEE*

***Abstract*—Hyperspectral anomaly detection (HAD) faces a fundamental challenge: background distributions vary significantly across scenes, making it difficult to generalize detection models without retraining. Most existing deep learning (DL) methods formulate HAD as a background reconstruction task, relying on reconstruction errors to assess anomalies and requiring scene-specific training data. This leads to high training costs and limited generalization. This work explores a fundamentally different question: can anomalies in hyperspectral images (HSIs) be detected using a model trained solely on natural images, without access to hyperspectral data or anomaly labels? To this end, we propose a hyperspectral anomaly detection framework based on pixel-wise likelihood estimation (HyLike), a new self-supervised framework that learns pixel-wise likelihood priors from natural grayscale images and transfers them to HSIs for anomaly detection without fine-tuning. This is made possible through a redesigned autoregressive generative network that models the conditional probability of each pixel given its spatial context. By learning pixel dependencies in natural images, the network provides a transferable prior whose pixel-wise likelihood can directly indicate anomaly intensity in HSIs. To bridge the modality gap between single-band grayscale images and HSIs, we apply independent component (IC) analysis to project HSIs into statistically ICs. Each component serves as a network input, from which we infer its anomaly significance and propose a self-adaptive fusion strategy to aggregate detection results across components. The resulting detection framework requires no labeled anomalies, retraining, or parameter adjustment. Extensive experiments on four benchmark HSIs demonstrate that HyLike generalizes effectively across diverse scenes, achieving competitive or superior performance compared to both traditional- and learning-based HAD methods. The code will be made publicly available at https://github.com/QingKangHe/HyLike**

***Index Terms*—Anomaly detection, hyperspectral image (HSI), natural image priors, pixel dependencies.**

Received 10 June 2026; accepted 3 July 2026. Date of publication 13 July 2026; date of current version 17 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42325104, Grant 42571420, and Grant 42271363. The work of Michael K. Ng was supported by the Department of Science and Technology of Guangdong Province (GDSTC): Guangdong and Hong Kong Universities "1 + 1 + 1" Joint Research Collaboration Scheme, National Key Research and Development Program of China under Grant 2024YFE0202900 and Grant RGC GRF 12300125. *(Corresponding author: Lina Zhuang.)*

Qingkang He is with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: heqingkang23@mails.ucas.ac.cn).

Lianru Gao and Lina Zhuang are with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: gaolr@aircas.ac.cn; zhuangln@aircas.ac.cn).

Michael K. Ng is with the Department of Mathematics, Hong Kong Baptist University, Hong Kong, China (e-mail: michael-ng@hkbu.edu.hk).

Yao Liu is with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of China, Beijing 100048, China (e-mail: liuyao@lasac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3712036

I. Introduction

HYPERSPECTRAL imaging technology excels at capturing detailed reflectance of ground objects through hundreds to thousands of electromagnetic spectrum bands, organized in a 3-D data cube that provides a novel approach to sense the world [1], [2]. The exceptional spectral resolution allows the pixels in the 3-D data cube to reflect the diagnostic spectral characteristics of materials, enabling diverse applications, including precision agriculture [3], mineral exploration [4], and national defense [5].

In many of these scenarios, the ability to detect unexpected or rare targets that deviate from the background is of critical importance. Motivated by this requirement, hyperspectral anomaly detection (HAD) aims to identify objects of interest, referred to as anomalies, that exhibit: 1) spectral deviations from their surrounding background distributions; and 2) low probability of occurrence, without requiring any prior information [6], [7]. In this context, anomalous targets typically occupy only a small number of pixels and exhibit spectral signatures that are distinct from the surrounding background. A representative example is the detection of fire hotspots following the terrorist attacks on the World Trade Center [8].

The HAD task shares conceptual similarities with the pixel-level classification task, with the key distinction that HAD resembles a binary classification problem with two predefined classes for each pixel: background and anomaly [9]. The high variability in imaging scenes and conditions results in complex data distributions in hyperspectral images (HSIs) [10], [11], [12], making HAD an exceptionally challenging task. To effectively separate the anomalies from background, researchers have developed numerous HAD methods falling into three main categories: 1) statistical-based methods; 2) traditional machine learning (ML)-based methods; and 3) deep learning (DL)-based methods.

A. Statistical-Based Methods

As the pioneering technique in HAD, statistical-based methods established the foundational framework for sub-

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

sequent developments. They assume that the background follows a certain statistical distribution, whereas anomalies deviate from this distribution. Thus, anomaly detection is performed by quantifying the degree of deviation of test pixels from the established background distribution. The benchmark statistical-based method is the Reed–Xiaoli (RX) algorithm [13] that assumes the testing pixel follows a multivariate Gaussian distribution under a binary hypothesis testing framework. Derived from the generalized likelihood ratio test (GLRT), the Mahalanobis distance (MD) between the testing pixel and the background distribution reflects the anomalous level. Depending on the method used to estimate the background distribution parameters, RX can be applied either to all pixels in the entire image (GRX) or within a local window (LRX) [8]. Later, to improve the performance of classical RX in complicated scenarios, many variants of RX have been proposed, such as kernel RX (KRX) [14], subspace RX (SSRX) [15], cluster-based RX (CBAD) [16], and weighted-RX (WRX) [17]. Considering that the fractional Fourier transform (FrFT) can better suppress noise and enhance the discrimination between background and anomalies, the fractional Fourier entropy (FrFE)-based method [18] uses FrFT as a preprocessing step followed by the RX detector in the optimal fractional domain. In addition, Liu et al. [19] proposed a two-step GLRT detector (2S-GLRT) to detect anomalies that occupy multiple pixels with unknown amplitudes. Chang employed a target-to-anomaly conversion mechanism [20] via the dummy variable trick to transform a series of existing hyperspectral target detection algorithms into their corresponding anomaly detection versions, achieving promising results. Subsequently, Chang et al. proposed iterative spectral–spatial [21], and cascaded spectral–spatial [22] HAD methods, which can improve detection performance for any given anomaly detector.

While the theory behind statistical-based methods is straightforward, the real-world background distribution is generally non-Gaussian, which fails to satisfy common assumptions. Besides, the anomaly detection performance achieved by second-order statistics is often suboptimal, as it may struggle to capture the high-level nonlinear features of HSIs.

B. Traditional ML-Based Methods

To alleviate the reliance on statistical assumptions, representation-based methods have emerged as promising approaches to address the limitations. Representation-based methods assume the background spectrum can be linearly represented by dictionary atoms, whereas anomalies cannot. The representation residual indicates the anomalous level of each pixel. Typical representation-based methods include sparse representation (SR) [23], [24], collaborative representation (CR) [25], [26], and low-rank representation (LRR) [27], [28]. The SR- and CR-based methods presume that the representation is competitive, i.e., ℓ_1 sparsity, and collaborative, i.e., ℓ_2 sparsity, respectively. Notable methods include background joint SR model (BJSR) [29], CR-based detector (CRD) [25], and others [30], [31], [32]. LRR-based methods decompose an HSI into low-rank and sparse components using preconstructed dictionaries. However, building a dictionary that adequately represents background variations while excluding anomalies remains challenging, prompting numerous studies to focus on learning pure dictionaries [28], [33]. Based on the rationale of low-rank methods, recent studies have further enhanced LRR by incorporating multiple complementary priors [34] and pixel segmentation strategies [35] to improve background modeling accuracy and computational efficiency. In contrast, matrix decomposition-based methods can directly decompose an HSI into its respective background and anomaly matrix [36]. LSDM-MoG [37] extracts the sparse matrix from data and models it as a mixture of Gaussian noise, achieving accurate detection under background noise contamination. From the 3-D perspective, tensor decomposition-based methods further consider the spatial and spectral properties of anomalous pixels [38]. In [39], a prior-based tensor approximation (PTA) is proposed, where spatial and spectral priors are integrated into a unified tensor decomposition framework.

Although traditional ML-based methods are not limited by statistical distributions, they often require manual design and careful tuning of the regularization parameters, which can be time-consuming and highly sensitive to the specific characteristics of the HSI.

C. DL-Based Methods

The paradigm of most existing DL-based methods is to harness the exceptional feature extraction ability of neural networks through self-supervised learning or semi-supervised learning to learn and reconstruct the background characteristics of the entire HSI, and then use the reconstruction error to identify anomalies. Xie et al. [40] leveraged clustering to identify background regions and trained a generative adversarial network (GAN) for image reconstruction, where anomalies in HSI show higher reconstruction errors compared with the background. Subsequently, various other semi-supervised learning-based methods have also emerged [41], [42], [43]. However, the background searching in semi-supervised learning-based methods is susceptible to anomaly contamination. If anomalous pixels are included in the training process, the network may learn the anomaly patterns to some extent, leading to a reduction in reconstruction error for anomalous pixels. This, in turn, can result in performance degradation. Some researchers turn to self-supervised learning for robust solutions. Auto-AD [44] introduces an adaptive-weighted loss function to focus the network's attention on reconstructing the background. Blind-spot network series, e.g., PDBSNet [45], BockNet [46], and DirectNet [47], impose a blind area in the receptive field of the network to reconstruct the center pixel using only the surrounding pixels. In addition, numerous methods such as GAED [48], GT-HAD [49], and DFAN-HAD [50] have been proposed to improve the reconstruction quality of the background, achieving desirable performance. Recently, Ma et al. [51] introduced the background suppression diffusion model (BSDM) for HAD, which treats the background as removable noise and applies existing HAD methods to the background-suppressed image. In addition, Xu et al. [52] proposed transformer-graph convolution-frequency (TransGCF), a unified framework that jointly exploits spatial, spectral, and frequency information to enhance anomaly-background separability. In addition, several studies [53], [54], [55] have explored model- and data-driven hybrid frameworks for HAD, unifying network training and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

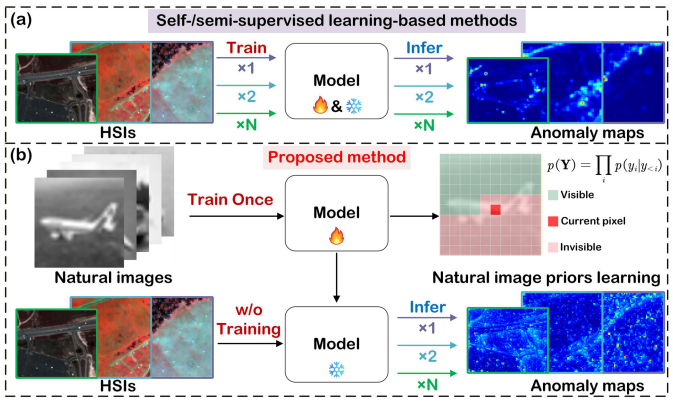


Fig. 1. Illustration of self- and semi-supervised learning-based methods and our HyLike. (a) Self- and semi-supervised learning-based methods require training the background reconstruction network for each HSI to fit its specific background distribution, which incurs high training costs; specifically, necessitating training the network N times for N individual detections. (b) Our proposed HyLike needs only training once on natural images to learn the natural image priors and transfer to HAD without (w/o) retraining or fine-tuning.

optimization-based problem solving within a single framework to improve interpretability.

Despite the promising performance of DL-based methods in feature extraction, their strong learning ability puts them at risk of mistakenly learning anomalous patterns as part of the background, resulting in low reconstruction errors for both background and anomalies. On the other hand, the diversity of imaging scenes and conditions necessitates training the network from scratch to learn the specific background distributions of each HSI before detection. This increases training costs and results in poor generalization, with stable performance achieved only on the training HSI.

D. Motivation and Contribution

As stated earlier, most existing DL-based methods use background reconstruction as a pretext task for HAD, where the reconstruction error depicts the anomalous level of pixels. However, as shown in Fig. 1, the specific background learning for each HSI causes these methods to be highly data-dependent and limits their generalization capability. To mitigate this limitation, considerable efforts have been devoted to transfer learning. Li et al. [56] proposed a transfer learning-based HAD method that learns a general spectral similarity model from a labeled reference HSI and detects on the HSI captured by the same sensor. Rao et al. [57] used paired pixels generated from labeled reference HSI to pretrain a Siamese network and then fine-tune it with coarse clustering results on the target HSI for anomaly detection. Despite their promising performance, the reliance on labeled reference HSIs remains a significant bottleneck for real-world HAD applications. Although some novel supervised learning-based methods start to train a general model on a large amount of HSIs with simulated anomalies, enabling only training once [58], [59], access to HSIs covering comprehensive scenes remains difficult. In addition, generating anomalous pixels that align with real-world anomaly distributions is relatively challenging, impacting subsequent detection performance [60].

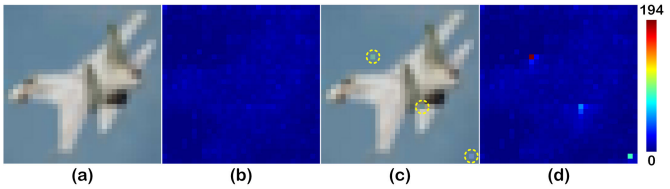


Fig. 2. Anomaly detection on an example RGB image based on pixel-wise likelihood modeling. (a) Original CIFAR-10 image. (b) Pixel-wise NLL of (a), predicted by a pretrained pixel-wise likelihood modeling network. (c) Anomaly-implanted version of (a) with the implanted anomaly positions highlighted by yellow circles. (d) Its corresponding pixel-wise NLL predicted by the same network. The pixel-wise likelihood modeling network can discern the anomalous pixels with very high NLL, which motivates us to explore the applications of likelihood modeling in HAD.

Natural images possess distinct pixel dependencies that are key to their structure [61]. In this article, natural images are defined as the photographic representations of the natural world within the visible light spectrum, along with their corresponding grayscale versions. The pixel dependencies in these natural images include short-range interactions that facilitate the reconstruction of a central pixel using neighboring pixels, and long-range dependencies that capture nonlocal similarities and textures. Moreover, these pixel dependencies exist not only in natural images but also extend to any image type sharing similar properties with natural images, e.g., HSIs. Notably, due to the atypical characteristics of anomalous pixels, the dependencies among background pixels are considerably stronger than those between anomalous and background pixels.

Inspired by this idea, we model pixel dependencies through pixel-wise likelihood modeling that factorizes the image probability density function (pdf) into a product of distributions conditioned on all previous pixels. The negative log-likelihood (NLL) of each pixel calculated from conditional probability directly quantifies the dependencies of its previous pixels; higher NLL indicates weaker pixel dependencies. As clearly shown in Fig. 2, the NLL of anomalous pixels is much higher than that of background pixels in the example RGB image, demonstrating their deviation from normal patterns of previous pixels.

From this novel perspective, we propose a hyperspectral anomaly detection framework based on pixel-wise likelihood estimation (HyLike) to address the previously stated limitations of training costs and poor generalization. HyLike implements likelihood modeling using a generative network, namely, a modified PixelCNN++ (mPixelCNN++), to capture pixel dependencies as natural image priors from grayscale natural images. Compared with the original single-channel PixelCNN++ [62], our mPixelCNN++ employs a shifted receptive field to avoid accessing multipixel anomalies, enhancing spatially contiguous anomaly detection. Furthermore, to apply this learned prior in the HAD task, independent component (IC) analysis (ICA) [63] is employed to bridge the modality gap while reducing the number of bands required for processing. An LRX-based weighted fusion strategy (LRX-WFS) is then introduced to highlight informative IC detection results using LRX-derived weights. In addition, an automatic ICA dimensionality reduction strategy (Auto-ICA), based on entropy-cosine similarity score (ECSS), is developed to automatically determine the optimal dimensionality, enabling

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

fully parameter-free detection. The theoretical innovations and significant contributions of this article can be summarized as follows:

1) We propose HyLike, a self-supervised HAD framework that transfers pixel-wise likelihood priors learned from natural grayscale images to HSIs. The prior is trained on the CIFAR-10 dataset using a modified PixelCNN++, without relying on any anomaly labels or hyperspectral data. To the best of our knowledge, this is the first work that leverages a network trained purely on natural images for HAD, and the first to introduce likelihood modeling into this field.
2) We redesign the autoregressive architecture and propose mPixelCNN++, which shifts the receptive field by excluding the adjacent upper and left neighbors from the prediction context. This shift weakens the direct dependency on adjacent pixels, thereby extending the model's sensitivity from isolated single-pixel anomalies to small spatially contiguous anomalies, consisting of only a few pixels.
3) To bridge the modality gap between natural grayscale images and hyperspectral data, we apply ICA to project HSIs into low-dimensional ICs. Each IC, represented as a single-band image, aligns with the source modality of the prior. We further introduce Auto-ICA, a task-specific, unsupervised strategy for automatically determining the optimal number of components by analyzing statistical patterns present in the anomaly maps—requiring no labels or manual tuning.
4) We design LRX-WFS, a self-adaptive fusion strategy that statistically evaluates the anomaly significance of each IC and assigns fusion weights accordingly. This approach robustly aggregates detection results across components while remaining entirely parameter-free.

The rest of this article is organized as follows. Section II introduces the pixel-wise likelihood modeling and transfer learning. Section III provides a detailed description of the methodology. Section IV shows the experimental results and analysis. Section V concludes the work of this article.

II. Preliminary

This section introduces the basic concepts of pixel-wise likelihood modeling and transfer learning.

A. Pixel-Wise Likelihood Modeling

Given an image $\mathbf{Y} \in \mathbb{R}^{n\times n}$, it can be unfolded row by row into a 1-D sequence $\mathbf{y} = [y_1, y_2, \ldots, y_{n^2}]$. The pixel-wise likelihood modeling aims to model the pdf of $\mathbf{Y}$ as the product of the conditional distributions by all its pixels y_i

$$p(\mathbf{Y}) = \prod_{i=1}^{n^2} p(y_i \mid y_{i-1}, y_{i-2}, \ldots, y_1) = \prod_{i=1}^{n^2} p(y_i \mid y_{<i}). \tag{1}$$

This formulation reflects an autoregressive structure where the current pixel y_i depends only on previously generated pixels. Fig. 3(a) and (b) further clarifies this conditional probability. The current pixel y_i (in red) is only related to its previous pixels (upper and left pixels in blue and green), while the "current" (in red) and "future" pixels (in pink) are invisible for y_i.

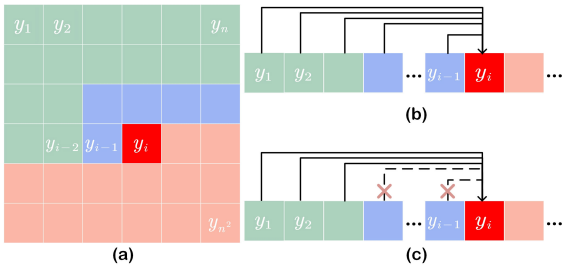


Fig. 3. Illustration of the conditional probability of an image. (a) Pixels relative to y_i are marked in red, blue, green, and pink. (b) Probability distribution of current pixel y_i (in red) only depends on its previous pixels (in blue and green). Both blue and green pixels are visible to the network, whereas pink and red pixels are not. (c) Probability distribution of the current pixel y_i only depends on nonadjacent upper left pixels (only green pixels), with blue pixels being excluded from consideration.

Based on its tractable and explicit autoregressive formulation, pixel-wise likelihood modeling has been widely adopted as a generative framework to model images in a pixel-by-pixel manner. In particular, PixelCNN++ [62] models pixel intensities using a mixture of logistic distributions, which provides a continuous and flexible density approximation. Compared with the discrete likelihood modeling adopted by the original PixelCNN [64] and gated PixelCNN [65], the use of continuous logistic mixture distributions enables more accurate and stable likelihood estimation. Thus, we adopt PixelCNN++ to model $p(y_i \mid y_{<i})$.

The PixelCNN++ model assumes that each pixel y_i is a discretized observation of a latent continuous variable v_i, drawn from a mixture of K logistic distributions

$$v_i \sim \sum_{k=1}^{K} \pi_{i,k} \cdot \text{Logistic}\left(\mu_{i,k}, s_{i,k}\right), \tag{2}$$

where $\pi_{i,k}$, $\mu_{i,k}$, and $s_{i,k}$ are the mixture weight, mean, and scale of the kth component for pixel i, respectively.

PixelCNN++ employs a CNN with a limited receptive field to restrict each pixel's dependency to its upper and left neighbors. Hence, this network, denoted as $f_{\boldsymbol{\theta}}(\cdot)$, achieves estimating parameters $\{\boldsymbol{\pi}_i, \boldsymbol{\mu}_i, \mathbf{s}_i\}$ of current pixel based on only its previous pixels $y_{<i}$

$$\{\boldsymbol{\pi}_i, \boldsymbol{\mu}_i, \mathbf{s}_i\} = f_{\boldsymbol{\theta}}\left(y_{<i}\right), \tag{3}$$

where $\boldsymbol{\pi}_i = \{\pi_{i,k}\}_{k=1}^{K}$, $\boldsymbol{\mu}_i = \{\mu_{i,k}\}_{k=1}^{K}$, and $\mathbf{s}_i = \{s_{i,k}\}_{k=1}^{K}$.

The conditional probability of y_i is then computed using the cumulative distribution function $F(y; \mu, s) = \sigma((y-\mu)/s)$ of the logistic distribution

$$p\left(y_i \mid \boldsymbol{\pi}_i, \boldsymbol{\mu}_i, \mathbf{s}_i\right) = \sum_{k=1}^{K} \pi_{i,k} \cdot \left[F\left(y_i + 0.5; \mu_{i,k}, s_{i,k}\right) - F\left(y_i - 0.5; \mu_{i,k}, s_{i,k}\right)\right], \tag{4}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

Finally, the NLL of pixel y_i is defined as

$$\text{NLL}\left(y_i\right) = -\log_2 p\left(y_i \mid \boldsymbol{\pi}_i, \boldsymbol{\mu}_i, \mathbf{s}_i\right). \tag{5}$$

This pixel-level NLL serves as an anomaly indicator, measuring how well the observed pixel aligns with the generative prior learned from natural images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

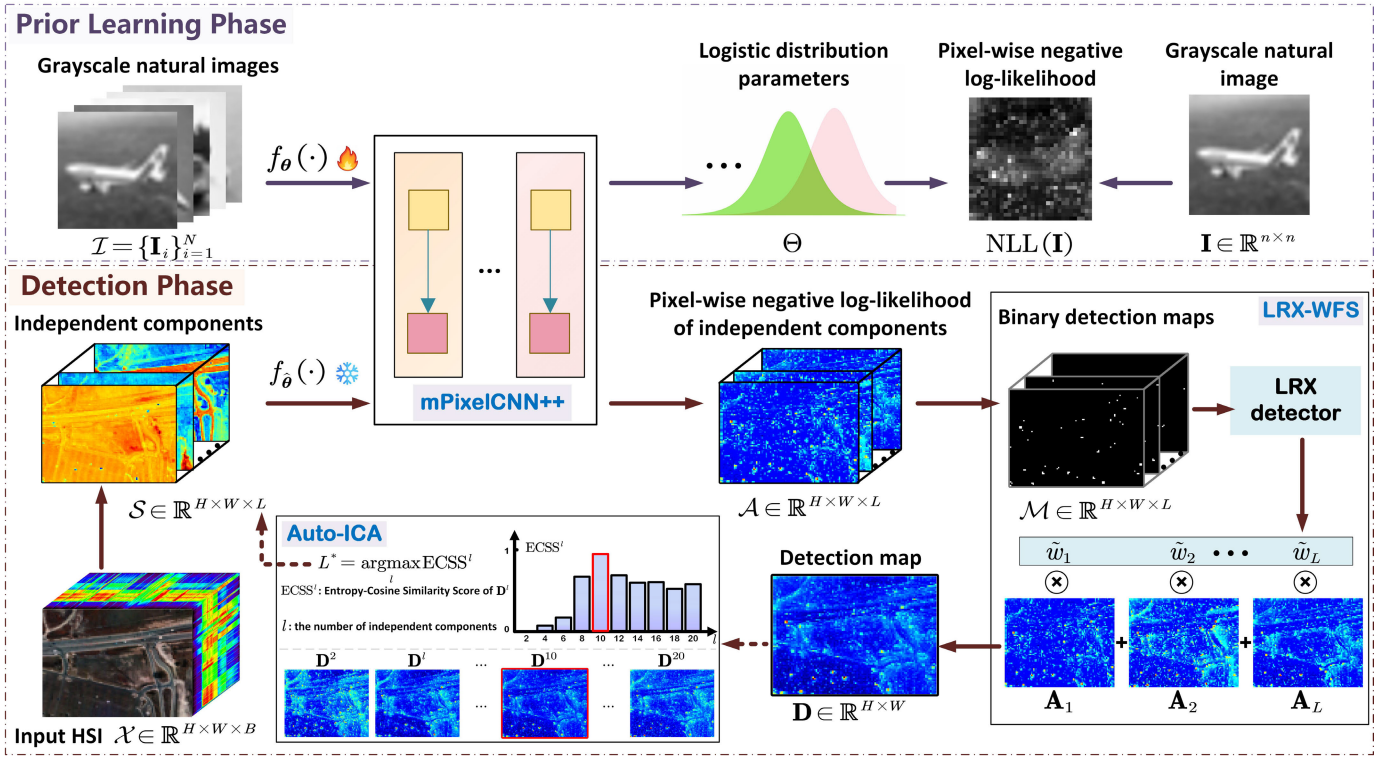


Fig. 4. Flowchart of the proposed HyLike.

B. Transfer Learning

Given the source domain $D_S = \{(x_{Si}, y_{Si})\}_{i=1}^{N_S}$, source task $\mathcal{T}_S$ with learning function $f_S(\cdot)$, and target domain $D_T = \{(x_{Ti}, y_{Ti})\}_{i=1}^{N_T}$, target task $\mathcal{T}_T$ with learning function $f_T(\cdot)$, where $x_{Si} \in \mathcal{X}_S$ and $x_{Ti} \in \mathcal{X}_T$ represent the instances of the source domain and target domain, respectively. y_{Si} and y_{Ti} represent the corresponding labels. Transfer learning enables the learning of the predictive function $f_T(\cdot)$ using the knowledge in D_S and $\mathcal{T}_S$ where $D_S \neq D_T$ or $\mathcal{T}_S \neq \mathcal{T}_T$ [66].

Specifically, we aim to use the grayscale of the natural images as the source domain D_n to learn the NLL predictive function $f_S(\cdot)$ and directly use it to estimate the pixel-wise NLL of HSIs or their features obtained by ICA transformation. Thus, the $\mathcal{T}_S$ and $\mathcal{T}_T$ are the same, i.e., predict the pixel-wise NLL for a single-band image. The HSIs that cover comprehensive scenes are hard to obtain, meaning that N_T is limited. In contrast, the natural image dataset is easily accessible and rich in content, i.e., N_S is relatively sufficient. Thus, we can learn $f_S(\cdot)$ well. We assume that the natural image domain D_n, HSI domain D_h, and its ICA feature domain D_f have approximately identical characteristics (more details in Section III-D), motivating us to conduct transfer directly $f_T(\cdot) \leftarrow f_S(\cdot)$.

III. Proposed HyLike

A. Overview

Fig. 4 depicts the overall framework of our proposed HyLike, which is divided into two phases as follows:

1) *Prior Learning Phase:* Our mPixelCNN++ is trained on grayscale natural images. The pixel-wise NLL calculated from the output of mPixelCNN++ reflects the dependencies between the current pixel and its nonadjacent upper left pixels.
2) *Detection Phase:* The input HSI first undergoes an ICA transformation to extract ICs. Subsequently, the pixel-wise NLL of ICs (NLLICs) is computed by leveraging mPixelCNN++ to infer each IC band individually. The detection map is then constructed by weighting and fusing the NLLICs, where the weights are assigned by our LRX-WFS. Finally, by traversing multiple ICA dimensionality, several detection maps are generated, among which the one with the highest ECSS is automatically selected as the optimal result by our Auto-ICA.

B. Problem Definition

Here, we give the problem definition and formulation. The 3-D HSI in the domain D_h is denoted as $\mathcal{X} \in \mathbb{R}^{H \times W \times B}$ where H, W, and B represent the height, width, and band number of the HSI, respectively. Let $\mathbf{X} \in \mathbb{R}^{B \times HW}$ represent the unfolding of $\mathcal{X}$ along the spectral dimension. The corresponding ICA feature of $\mathcal{X}$ in the domain D_f is denoted as $\mathcal{S} \in \mathbb{R}^{H \times W \times L}$, where L is the number of ICs preserved after ICA transformation. $\mathcal{I} = \{\mathbf{I}_i\}_{i=1}^{N}$ denotes the grayscale natural image containing N images. $\mathbf{I}_i \in \mathbb{R}^{n \times n}$ belongs to the natural image domain D_n.

Although HSIs involve both spectral and spatial dependencies, HyLike handles them in a decoupled manner. ICA first projects the original spectral correlations into statistically IC maps, where spectrally distinctive anomalies can be highlighted in the transformed feature space. We then use dataset $\mathcal{I}$ and mPixelCNN++ to learn an NLL predictive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

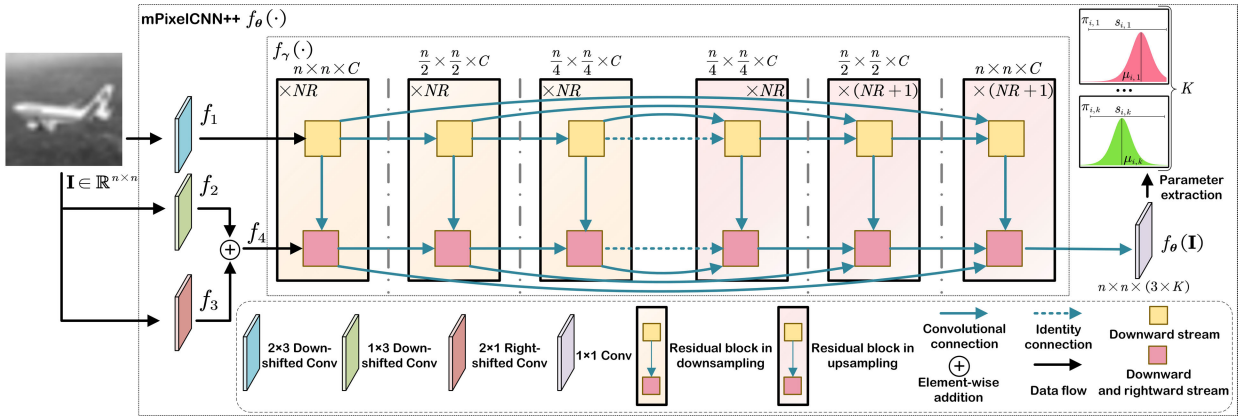


Fig. 5. Architecture of our mPixelCNN++. It closely follows the original PixelCNN++ framework, except that the receptive fields of feature maps $f_i, i = 1, 2, 3, 4$ are confined by different parameter settings in the DSC and RSC, enabling it to predict the NLL of each pixel based on its nonadjacent upper left pixels. It adopts a U-Net-like structure and composes three distinct resolution levels, each has NR or $NR + 1$ residual blocks, with convolutional and identity connections. The number of channels in the feature maps is denoted by C. For each pixel y_i, the corresponding output in $f_{\boldsymbol{\theta}}(\mathbf{I})$ contains K logistic distributions, each parameterized by a weight $\pi_{i,k}$, mean $\mu_{i,k}$, and scale $s_{i,k}$. The NLL will be calculated for anomaly detection upon extracting these parameters for each pixel y_i.

function $f_{\boldsymbol{\theta}}(\cdot)$, which is applied to each IC to model pixel-wise spatial conditional probabilities. The resulting NLL reflects the local conditional statistical dependency of each pixel, where a higher NLL indicates weaker pixel dependencies and thus a higher anomalous level. Therefore, HyLike does not rely on high-level semantic object knowledge from natural images, but uses the pixel-wise likelihood modeling capability learned from natural images as the transferred prior for anomaly detection.

C. Model

This section introduces the mPixelCNN++ architecture and the corresponding training and inference process.

1) Architecture: As shown in Fig. 5, our mPixelCNN++ has a U-Net-like structure comprising two downsampling and upsampling steps, resulting in three distinct resolution levels. All three resolution levels have the same number of channels C but differ in the number of residual blocks. The same resolution levels in downsampling and upsampling are connected through convolutional layers. After a 1×1 convolution operation, the output from the final downward and rightward stream includes the distribution parameters for all pixels, denoted as $f_{\boldsymbol{\theta}}(\mathbf{I})$, whose dimensionality is $n \times n \times (3 \times K)$, where K is the number of logistic distributions.

Our mPixelCNN++ adheres to the original PixelCNN++ architecture, except that we modify the initial operations of the network, i.e., the receptive fields of the feature maps f_i, $i = \{1, 2, 3, 4\}$ are further shifted by different parameter settings in the downshifted convolution (DSC) and rightshifted convolution (RSC). Fig. 6 demonstrates the above processing. The grayscale image $\mathbf{I}$ is initially zero-padded. This padded image is then concurrently processed by three $k \times j$ convolution kernels. The padding operation $\text{Padding}_{l,r,t,b}$, applied to the left (l), right (r), top (t), and bottom (b) sides of $\mathbf{I}$, is designed to align with the kernel sizes k and j, and is detailed as follows. For the two DSCs, the padding is $l = r = (j - 1)/2$, $t = k - 1$, and $b = 0$. For the RSC, the padding is $l = j - 1$, $t = k - 1$, and $r = b = 0$. After being processed by 2×3, 1×3, and 2×1 convolution kernels,

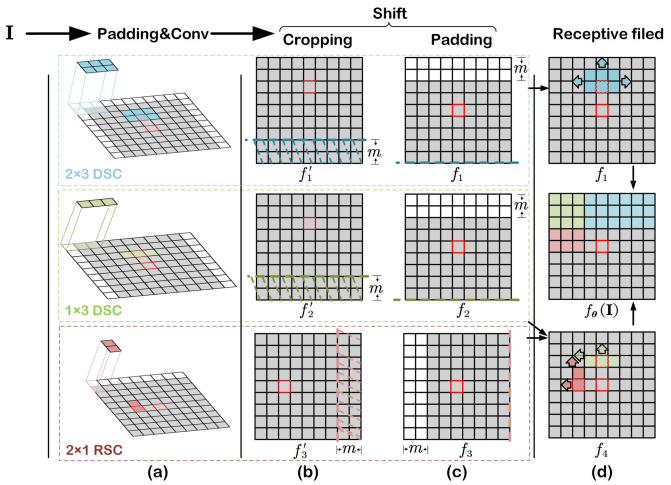


Fig. 6. Illustration of shift operation in receptive field. (a) Padding and convolution operation in DSC and RSC. The padding operation aims to maintain the height and width of feature maps. (b) and (c) Shift operation applied to feature maps f_1', f_2', and f_3', involving the cropping of the last m rows (or columns) from DSC (or RSC) feature maps, followed by padding zeros in the first m rows (or columns). The shift operation enables the central pixels (the red rectangles) in f_1, f_2, and f_3 to correspond to the red dashed rectangles in f_1', f_2', and f_3', respectively, resulting in the receptive field in (d) where the gray rectangles are invisible to the network. The blue, red, and green arrows indicate the expanding directions of the receptive field in the subsequent forward propagation through the network.

we obtain the feature maps f_1', f_2', and f_3'. Subsequent shift operations are conducted as follows:

$$f_i = \text{Padding}_{0,0,m,0}\left(f_i'(1:n-m, 1:n)\right), \quad i = 1, 2, \tag{6}$$

$$f_3 = \text{Padding}_{m,0,0,0}\left(f_3'(1:n, 1:n-m)\right), \tag{7}$$

$$f_4 = f_2 + f_3, \tag{8}$$

where m represents the shift displacement, determining the step size for both the cropping and padding operations. The notation $f'(a:b, c:d)$ corresponds to a cropping operation, signifying the extraction of elements from a matrix or tensor f', spanning rows from a to b and columns from c to d. After the shift operation, the receptive field of the central pixel is shifted. For two DSCs, the central pixel (the red rectangle)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

of f_i, where $i = 1, 2$, corresponds to the mth pixel above it, i.e., the red dashed rectangle in f_i'. For RSC, the central pixel corresponds to the mth pixel to the left of it. Finally, the receptive fields of f_1 and f_4 are shown in Fig. 6(d). In our HyLike framework, we set $m = 2$, whereas the original PixelCNN++ uses a default displacement of $m = 1$.

The subsequent residual blocks, denoted as $f_{\boldsymbol{\gamma}}(f_1, f_4)$, follow the original PixelCNN++ architecture,[1] where the receptive field of f_1 will expand upward (blue arrows). In contrast, f_4 will expand toward the upper left (red and green arrows), effectively confining the final receptive field $f_{\boldsymbol{\theta}}(\mathbf{I})$ to nonadjacent upper left pixels [shown in Figs. 3(c) and 6(d)]. To simplify notation, we denote the output of the model as

$$\Theta = f_{\boldsymbol{\theta}}(\mathbf{I}) = f_{\boldsymbol{\gamma}}(f_1, f_4), \tag{9}$$

where $\Theta = \{\boldsymbol{\pi}_i, \boldsymbol{\mu}_i, \mathbf{s}_i\}_{i=1}^{n^2}$ represents the collection of logistic mixture parameters for all pixels in $\mathbf{I}$. To obtain the parameters for a specific pixel y_i, a parameter extraction process is performed on Θ, retrieving its corresponding mixture weights, means, and scales: $\{\pi_{i,k}, \mu_{i,k}, s_{i,k}\}_{k=1}^{K}$.

2) Training and Inference: The likelihood modeling aims to minimize the NLL of the image $\mathbf{I}$. Naturally, the loss function $\mathcal{L}(\mathbf{I}; \boldsymbol{\theta})$ is derived

$$\mathcal{L}(\mathbf{I}; \boldsymbol{\theta}) = \sum_{y_i \in \mathbf{I}} \mathrm{NLL}(y_i), \tag{10}$$

where the $\mathrm{NLL}(y_i)$ is defined in (5).

Once finishing the training, we obtain our mPixelCNN++ $f_{\hat{\boldsymbol{\theta}}}$ with fixed network parameters $\hat{\boldsymbol{\theta}}$. The inference produces the distribution parameters of each pixel for a single-band image $\mathbf{I}$. Subsequently, we can calculate the pixel-wise NLL of each pixel in $\mathbf{I}$, and this 2-D NLL matrix is denoted as $\mathrm{NLL}(\mathbf{I}) \in \mathbb{R}^{n \times n}$.

D. Domain Transfer

In this section, we explore the foundational assumptions supporting the transfer of mPixelCNN++ from the natural image domain to the HSI and ICA feature domains.

Our key assumption for anomaly detection is that natural images possess distinct pixel dependencies, manifested as strong correlations among pixels. We hypothesize that such pixel dependencies exist not only in the natural image domain (D_n) but also in the HSI domain (D_h) and its ICA-transformed feature domain (D_f). To validate this assumption, we compare the pixel value $\mathbf{I}(i, j)$ at position (i, j) with those of its neighboring pixels $(i - m, j)$ and $(i, j - m)$ across the domains mentioned above. The strength of this pixel dependency is quantified using the coefficient of determination (R^2), where a higher value indicates stronger linear pixel-wise correlation. The 20 single-band images are randomly selected from the training set of each domain, including HAD100, CIFAR-10, and the ICA-transformed domain derived from the HAD100 training set, to perform this analysis (see Section IV-A for more details about the datasets). The results are visualized in Fig. 7, showing that both horizontal and vertical pixel dependencies exhibit a strong linear relationship for all these domains, with R^2 ranging from 0.53 to 0.86. As the shift displacement m increases from 1 to 2, R^2 decreases across all domains, indicating weaker yet still notable dependencies. These consistently shared structural characteristics across domains justify our assumption that the pixel-wise likelihood priors learned from natural images can be effectively transferred to the HSI domain and its ICA feature space.

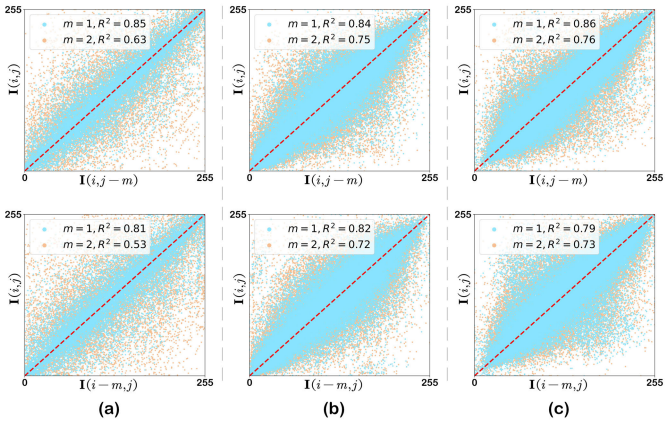


Fig. 7. Scatter plots showing horizontal (first row) and vertical (second row) pixel dependencies. (a) Natural image domain (D_n). (b) HSI domain (D_h). (c) ICA-transformed feature domain (D_f). The pixel displacement is denoted by m, and R^2 indicates the coefficient of determination.

E. ICA Dimensionality Reduction and Initial Detection

The mPixelCNN++ network is trained on natural grayscale images, leading to a modality gap when applied to HSIs. Directly feeding HSIs band-by-band into the network incurs a high computational cost, as it requires modeling the distribution of each pixel in each band. However, strong interband correlations in HSIs allow their spectral vectors to be well approximated in a low-dimensional subspace [67]. Thus, we first apply dimensionality reduction and then feed the resulting components into the network to mitigate the modality gap and reduce computation.

Common dimensionality reduction methods for HSI include the linear mixture model (LMM) [68], principal component analysis (PCA), and ICA. We adopt ICA for two main reasons:

1) PCA and ICA generally preserve informative signals better than LMM. The LMM decomposes an HSI into an endmember matrix (i.e., spectra of pure materials) and the corresponding abundance maps (i.e., the proportion of each material in each pixel). Although most endmember extraction algorithms are designed to be noise-robust [69], they may mistakenly treat spectral anomalies as noise and exclude them during decomposition, leading to loss of target-related information.
2) Unlike PCA, which only enforces linear decorrelation, ICA seeks statistical independence, making it better at separating physically meaningful classes (e.g., background or anomalous targets). In other words, ICA tends to separate anomalous and background materials into different components, a property that PCA lacks.

The procedure for reducing HSI dimensionality via ICA and inputting the resulting components into the pretrained mPixelCNN++ network is described as follows. For the unfolding HSI $\mathbf{X} \in \mathbb{R}^{B \times HW}$, we first center and whiten it to meet

[1] https://github.com/pclucas14/pixel-cnn-pp

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

the requirements for ICA. In the whitening process, we apply dimensionality reduction to remain L bands in $\mathbf{X}' \in \mathbb{R}^{L\times HW}$

$$\left\{\mathbf{U}, \boldsymbol{\sigma}, \mathbf{V}^T\right\} = \mathrm{SVD}_L\left(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{x}}\right), \tag{11}$$

$$\mathbf{X}' = \left(\frac{\mathbf{U}}{\boldsymbol{\sigma}}\right)^T \times \left(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{x}}\right) \times \sqrt{HW}, \tag{12}$$

where $\boldsymbol{\mu}_{\mathbf{x}}$ is the mean vector of $\mathbf{X}$ across the sample dimension. $\mathrm{SVD}_L(\cdot)$ is a singular value decomposition function returning a subset of singular values and vectors. The subscript L indicates that the function returns only the singular vectors corresponding to the L largest singular values, i.e., $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_L]$, $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_L]$, and $\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \ldots, \sigma_L]$. The division $\mathbf{U}/\boldsymbol{\sigma}$ equals $[\mathbf{u}_1/\sigma_1, \mathbf{u}_2/\sigma_2, \ldots, \mathbf{u}_L/\sigma_L]$. We then use $\mathbf{X}'$ as the mixed signal matrix. Through FastICA [70] and folding the derived ICs back to 3-D format, the ICA feature of HSI $\mathcal{S} \in \mathbb{R}^{H\times W\times L}$ is obtained. Subsequently, each individual IC $\mathbf{S}_l$ is fed into our mPixelCNN++ to obtain the corresponding negative log-likelihood map $\mathbf{A}_l$

$$\mathbf{A}_l = \mathrm{NLL}\left(\mathbf{S}_l\right), \quad l = 1, 2, \ldots, L, \tag{13}$$

where $\mathrm{NLL}(\cdot)$ denotes the pixel-wise NLL function previously defined in (5). Here, it is applied element-wise to every pixel in $\mathbf{S}_l$, producing an output map $\mathbf{A}_l$ of the same size. Symbol $\mathcal{A} \in \mathbb{R}^{H\times W\times L}$ denotes the NLLICs. $\mathbf{A}_l$ and $\mathbf{S}_l$ denote the lth bands of $\mathcal{A}$ and $\mathcal{S}$, respectively.

F. LRX-Based Weighted Fusion Strategy

When the initial detection results (i.e., each band of $\mathcal{A}$) are acquired, it is necessary to weight and fuse them to construct the final detection map $\mathbf{D} \in \mathbb{R}^{H\times W}$. The reason is that not all initial detection results are relevant or helpful in reflecting anomaly features. For instance, some ICs may represent noise, while others contribute differently to anomaly detection, as each IC may correspond to different ground objects. Therefore, a weighting mechanism is necessary to balance their contributions.

Given the multidimensional initial detection results $\mathcal{A}$ obtained from ICA components, we adopt an LRX-WFS to integrate these results and produce the final detection map $\mathbf{D}$. The fusion process consists of the following key steps:

First, for each anomaly map $\mathbf{A}_l$ $(l = 1, \ldots, L)$ of each ICA component, we compute a binary mask $\mathbf{M}_l \in \mathbb{R}^{H\times W}$ using an adaptive thresholding method inspired by [71]. Specifically, the threshold δ_l is computed as

$$\delta_l = u_l + (M_l - u_l)\sqrt{\frac{u_l}{M_l}}, \tag{14}$$

where u_l and M_l denote the mean and maximum value of $\mathbf{A}_l$, respectively. A pixel in $\mathbf{A}_l$ is set to 1 in $\mathbf{M}_l$ if its value exceeds δ_l, and 0 otherwise.

Next, we construct the binary mask tensor $\mathcal{M} \in \mathbb{R}^{H\times W\times L}$ with $\mathcal{M}(:, :, l) = \mathbf{M}_l$, and use it to compute a band-wise importance weight vector $\mathbf{w} = [w_1, w_2, \ldots, w_L]^T \in \mathbb{R}^L$ by correlating each binary mask $\mathbf{M}_l$ with the LRX detection map $\mathbf{Z} \in \mathbb{R}^{H\times W}$. Each weight w_l is calculated as

$$w_l = \frac{\mathrm{sum}\left(\mathbf{M}_l \odot \mathbf{Z}\right)}{\mathrm{sum}\left(\mathbf{M}_l\right)}, \quad l = 1, \ldots, L, \tag{15}$$

where $\odot$ denotes the element-wise multiplication and $\mathrm{sum}(\cdot)$ represents the summation over all elements.

To ensure numerical stability, the weight vector is normalized using min–max normalization

$$\tilde{\mathbf{w}} = \mathrm{normalize}\left(\mathbf{w}\right) = \frac{\mathbf{w} - \mathbf{w}_{\min}}{\mathbf{w}_{\max} - \mathbf{w}_{\min}}, \tag{16}$$

where $\mathrm{normalize}(\cdot)$ denotes a min–max normalization function. $\mathbf{w}_{\min}$ and $\mathbf{w}_{\max}$ denote the minimum and maximum values of $\mathbf{w}$, respectively.

Finally, the fused anomaly map $\mathbf{D}$ is obtained by weighted summation

$$\mathbf{D} = \sum_{l=1}^{L} \tilde{w}_l \times \mathbf{A}_l. \tag{17}$$

The LRX detection map $\mathbf{Z}$ used above is computed based on a local background estimation. Specifically, for each pixel $\mathbf{x}_i$, we define an inner window of size $(m + 1)$ and an outer window of size $(m + 3)$ to estimate the local background mean $\boldsymbol{\mu}_{b_i}$ and covariance $\boldsymbol{\Sigma}_{b_i}$. The LRX anomaly score z_i is then computed as

$$z_i = \left(\mathbf{x}_i - \boldsymbol{\mu}_{b_i}\right)^T \boldsymbol{\Sigma}_{b_i}^{-1}\left(\mathbf{x}_i - \boldsymbol{\mu}_{b_i}\right), \quad i = 1, \ldots, HW. \tag{18}$$

To reduce computational cost, we only compute z_i for pixel positions where at least one $\mathbf{M}_l = 1$, significantly accelerating the LRX process compared to evaluating it over the entire image.

G. Auto-ICA

The number of ICs influences the detection performance in two aspects. Underestimating the number of ICs leads to the loss of informative content in HSI, resulting in suboptimal separation of background and anomaly. Conversely, overestimating the number of ICs produces noise components and computational burdens on network inference. In real-world applications, the prior of the number of ICs is typically unavailable and must be manually tuned based on empirical experience.

To make HyLike fully parameter-free, we propose **Auto-ICA**, an unsupervised strategy that automatically estimates the optimal ICA dimensionality by evaluating the quality of resulting detection maps. The key idea is that desirable detection maps tend to satisfy two properties: they exhibit a clear foreground–background separation, reflected by low entropy, and they are structurally consistent with detection maps obtained at nearby ICA dimensionalities, reflected by high cosine similarity.

Let $\{\mathbf{D}^l\}$ denote the detection maps generated by traversing the number of ICs $l \in \{2, 4, \ldots, 20\}$ with an interval of 2. For each $\mathbf{D}^l$, we evaluate its quality using two complementary measures: entropy value and cosine similarity.

We first compute the entropy value H^l. To ensure valid entropy computation, each $\mathbf{D}^l$ is first discretized to an 8-bit grayscale image by linear min–max normalization and quantization into 256 levels. The Shannon entropy is then calculated based on its histogram distribution

$$H^l = -\sum_i p_i^l \log p_i^l, \tag{19}$$

where p_i^l is the normalized frequency of intensity level i in the 8-bit version of $\mathbf{D}^l$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

Next, we compute the cosine similarity score (CSS) to quantify how structurally consistent $\mathbf{D}^l$ is with detection maps from other IC numbers. It is calculated as the average cosine similarity between $\mathbf{D}^l$ and all other detection maps $\mathbf{D}^j (j \neq l)$

$$\mathrm{CSS}^l = \frac{1}{9} \sum_{j \neq l} \frac{\mathrm{vec}\left(\mathbf{D}^l\right)^T \mathrm{vec}\left(\mathbf{D}^j\right)}{\|\mathbf{D}^l\|_2 \cdot \|\mathbf{D}^j\|_2}, \tag{20}$$

where $\mathrm{vec}(\cdot)$ denotes the vectorization into a 1-D column vector.

We then collect all entropy values and CSSs into column vectors, i.e., $\mathbf{H} = [H^2, H^4, \ldots, H^{20}]^T$ and $\mathbf{CSS} = [\mathrm{CSS}^2, \mathrm{CSS}^4, \ldots, \mathrm{CSS}^{20}]^T$. To align both measures with the convention that higher values indicate better detection quality, we apply min–max normalization to obtain the final scores

$$\widetilde{\mathbf{ES}} = 1 - \mathrm{normalize}\left(\mathbf{H}\right), \quad \widetilde{\mathbf{CSS}} = \mathrm{normalize}\left(\mathbf{CSS}\right), \tag{21}$$

where $\widetilde{\mathbf{ES}}$ denotes an entropy-based score vector, $\mathrm{normalize}(\cdot)$ denotes the min–max normalization function defined in (16).

The final ECSS for each l is given by

$$\mathrm{ECSS}^l = \widetilde{\mathrm{ES}}^l \cdot \widetilde{\mathrm{CSS}}^l, \tag{22}$$

where $\widetilde{\mathrm{ES}}^l$ and $\widetilde{\mathrm{CSS}}^l$ are elements from $\widetilde{\mathbf{ES}}$ and $\widetilde{\mathbf{CSS}}$, respectively.

Finally, the optimal ICA dimensionality L^* is selected as the one that maximizes ECSS

$$L^* = \arg\max_l \mathrm{ECSS}^l. \tag{23}$$

In our experiments, we search l from 2 to 20 with an interval of 2. The lower bound assumes the presence of at least two distinct sources (background and anomaly), while the upper bound ensures sufficient representational capacity. The interval of 2 offers a good balance between performance and computational cost, as detection maps from nearby IC numbers often exhibit similar patterns.

The proposed HyLike is summarized in Algorithms 1 and 2. It is worth noting that the prior learning phase only requires training once, and the detection phase operates without the need for manually set parameters, such as the number of ICA components.

Algorithm 1 Prior Learning Phase of HyLike

Input: Grayscale natural image dataset $\mathcal{I} = \{\mathbf{I}_i\}_{i=1}^N$, number of residual blocks *NR*, number of feature channels C, and logistic mixture components K.
Initialization: Assign the shift displacement $m = 2$.
Output: The mPixelCNN++ with the learned parameters $\hat{\boldsymbol{\theta}}$.
for each batch in $\mathcal{I}$ **do**
1: Network forward using (6)∼(9);
2: Calculate the loss using (10);
3: Network backward;
end for

Algorithm 2 Detection Phase of HyLike

Input: HSI $\mathcal{X} \in \mathbb{R}^{H \times W \times B}$, the learned prior $f_{\hat{\boldsymbol{\theta}}}$.
Initialization: Detection map series $\{\mathbf{D}^l\}$ with zeros.
Output: the final detection map $\mathbf{D}^{L^*}$.
for $l = 2$ **to** 20 in step of 2 **do**
1: Acquire the ICA features $\mathcal{S} \in \mathbb{R}^{H \times W \times L}$ using (11), (12), and FastICA;
2: Obtain the initial detection maps $\mathcal{A} \in \mathbb{R}^{H \times W \times L}$ using (13);
3: Weight and fuse $\mathcal{A}$ to obtain the detection map $\mathbf{D}^l$ using (14)∼(18);
end for
4: Calculate the entropy-cosine similarity score ECSS^l using (22);
5: Select the optimal dimensionality L^* for reduction using (23) and corresponding final detection map $\mathbf{D}^{L^*}$;

IV. Experimental Results and Analysis

A. Datasets

1) CIFAR-10 Dataset [72]: This natural image dataset[2] comprises 60 000 color images of a size 32×32, categorized into ten classes, including airplane, bird, cat, and others. We use the conversion, i.e., $\mathbf{I} = R \times 0.2989 + G \times 0.5870 + B \times 0.1140$, to transform RGB images to grayscale for training our network.

2) HSI Dataset: The experiments were conducted on four real-world HSIs. The first two were collected using the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over an urban area in California. The first image HYDICE-I [73][3] has 21 anomaly targets [74], while the second one HYDICE-II [6][4] has 10 anomaly targets [71], all of which are vehicles. The last two HSIs (HAD100-I and HAD100-II), collected by the Airborne Visible/Infrared Imaging Spectrometer-Next Generation (AVIRIS-NG), are a subset of the HAD100 dataset [58],[5] which primarily features compact manufactured objects, such as vehicles, boats, and buildings. For these two scenes, HAD100-I and HAD100-II include 12 and 25 anomaly targets, respectively. Table I details the aforementioned HSI dataset, including image size and spatial resolution. The corresponding pseudo-color images and ground-truth maps are depicted in Fig. 8. Notably, consistent with the guidelines in [75], [76], [77], and [78], our primary evaluation strictly utilizes real-world datasets with small anomalies, avoiding synthetic data and easily identifiable large targets.

TABLE I
Characteristics of the Four HSIs

Dataset	Image Size	Resolution	Spectral Bands	Location	Sensor
HYDICE-I	80×100	1 m	162	California	HYDICE
HYDICE-II	60×80	1 m	162	California	HYDICE
HAD100-I	64×64	8.4 m	276	\	AVIRIS-NG
HAD100-II	64×64	3 m	276	\	AVIRIS-NG

[2]CIFAR-10 dataset website: https://www.cs.toronto.edu/∼kriz/cifar.html
[3]HYDICE-I dataset website: https://github.com/zephyrhours/Hyperspectral-Anomaly-Detection-SLW_LRRSTO-MLW_LRRSTO-
[4]HYDICE-II dataset website: https://github.com/DegangWang97/IEEE_TGRS_BS3LNet
[5]HAD100-I and HAD100-II dataset website: https://zhaoxuli123.github.io/HAD100/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

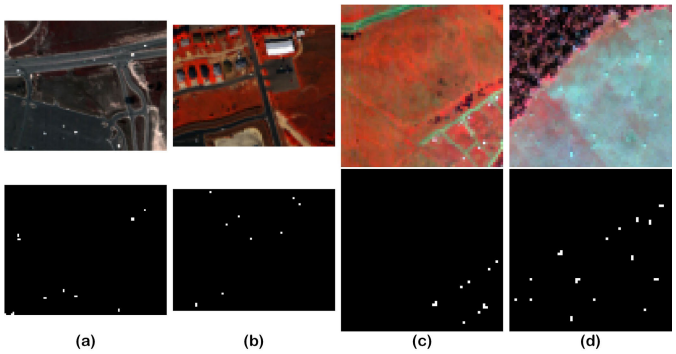


Fig. 8. Pseudocolor images and corresponding ground-truth maps of the four hyperspectral scenes. (a) HYDICE-I. (b) HYDICE-II. (c) HAD100-I. (d) HAD100-II.

B. Experimental Setup

1) Evaluation Metrics: To comprehensively evaluate the experimental methods, we adopt the 3-D receiver operating characteristic (ROC) curve [79] and the area under the curve (AUC) as quantitative metrics. Specifically, the 3-D ROC curve is employed to characterize the relationship between the probability of detection (PD), the false alarm rate (FAR), and the threshold τ. The 3-D ROC curve allows for the production of the 2-D ROC curve of (PD, FAR), (PD, τ), and (FAR, τ). Furthermore, based on the 3-D ROC curve, eight AUC values can be obtained [77], [80], providing a more comprehensive evaluation of detection performance. Equation (24) gives the definition of metrics mentioned above

$$\begin{cases} \mathrm{PD}(\tau) = \dfrac{\mathrm{TP}(\tau)}{N_{\mathrm{ANO}}}, \mathrm{FAR}(\tau) = \dfrac{\mathrm{FP}(\tau)}{N_{\mathrm{BKG}}}, \\ \mathrm{AUC}_{\mathrm{ADP}} = \mathrm{AUC}_{(\mathrm{D},\tau)} = \displaystyle\int_0^1 \mathrm{PD}(\tau)\, d\tau, \\ \mathrm{AUC}_{(\mathrm{F},\tau)} = \displaystyle\int_0^1 \mathrm{FAR}(\tau)\, d\tau, \\ \mathrm{AUC}_{\mathrm{BDP}} = 1 - \mathrm{AUC}_{(\mathrm{F},\tau)}, \\ \mathrm{AUC}_{(\mathrm{D,F})} = \displaystyle\int_0^1 \mathrm{PD}(\mathrm{FAR}) d(\mathrm{FAR}), \\ \mathrm{AUC}_{\mathrm{JAD}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{\mathrm{ADP}}, \mathrm{AUC}_{\mathrm{JBS}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{\mathrm{BDP}}, \\ \mathrm{AUC}_{\mathrm{SNPR}} = \dfrac{\mathrm{AUC}_{(\mathrm{D},\tau)}}{\mathrm{AUC}_{(\mathrm{F},\tau)}}, \mathrm{AUC}_{\mathrm{ADBS}} = \mathrm{AUC}_{\mathrm{ADP}} + \mathrm{AUC}_{\mathrm{BDP}}, \\ \mathrm{AUC}_{\mathrm{OADP}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{\mathrm{ADP}} + \mathrm{AUC}_{\mathrm{BDP}}, \end{cases} \tag{24}$$

where $\mathrm{TP}(\tau)$, $\mathrm{FP}(\tau)$, N_{ANO}, and N_{BKG} represent the number of detected anomalous pixels with a certain threshold τ, the number of false positive pixels, the total number of anomalous pixels, and the total number of background pixels, respectively. In practice, the thresholds τ are taken from the distinct values of the detector's response on the data sample. Following (7) in [79], the metrics $\mathrm{AUC}_{(\mathrm{D},\tau)}$, $\mathrm{AUC}_{(\mathrm{F},\tau)}$, and $\mathrm{AUC}_{(\mathrm{D,F})}$ are computed by using the normalized detector values as thresholds and performing trapezoidal integration. To obtain correct AUC values, we apply normalization such that the ROC curve passes through (0, 0) and (1, 1); the normalized AUC and the corresponding normalized ROC curve are computed using the code provided in [79, Sec. VI-E]. Metrics $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{\mathrm{ADBS}}$, and $\mathrm{AUC}_{\mathrm{OADP}}$ are adopted to assess the overall performance of a detector. Metrics $\mathrm{AUC}_{\mathrm{ADP}}$ and $\mathrm{AUC}_{\mathrm{JAD}}$ measure the detection capability of a detector. Metrics $\mathrm{AUC}_{\mathrm{BDP}}$, $\mathrm{AUC}_{\mathrm{JBS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$ indicate the capability of suppressing background responses [48]. In addition, we use the colorized detection map and box–whisker plot to assess the performance qualitatively. The detector responses of the two categories (ground-truth background and anomaly pixels) are collected and visualized using two separate boxes (blue for the background and red for the anomaly). Specifically, the minimum, 10th percentile, median, 90th percentile, and maximum values are used to define the lower whisker, box bottom, median line, box top, and upper whisker of each box–whisker plot, respectively, where the box represents the [10%, 90%] statistical range [44], [47].

2) Comparison Methods: We compare our HyLike with 12 representative methods, including statistical (GRX [13], LRX [8], and 2S-GLRT [19][6]), traditional ML (FrFE [18][7] and KIFD [81][8]), and DL-based approaches (GAED [48],[9] Auto-AD [44],[10] GT-HAD [49],[11] DFAN-HAD [50],[12] AETNet [58],[13] BSDM [51],[14] and TransGCF [52][15]). Among these DL-based methods, AETNet is trained on a large-scale hyperspectral dataset with simulated anomalies and only requires training once. BSDM leverages the reverse diffusion process to suppress background components, followed by applying a conventional HAD detector (RX in our experiments) on the processed data. While the other DL-based methods follow the self-supervised learning-based paradigm, using background reconstruction error to identify anomalies. It is worth noting that our HyLike is the only model trained on natural images without any further fine-tuning on HSI data.

3) Implementation Details: We adopt the default training/testing split of CIFAR-10, with 50 000 training and 10 000 test images. The number of residual blocks NR is set to 5. The number of feature channels C is set to 160. The mixture includes 10 logistic components ($K = 10$). We use the Adam optimizer with an initial learning rate of 10^{-3}, decayed to 10^{-6} over 200 epochs via a half-cycle cosine decay schedule. The batch size is 128. All CIFAR-10 images are converted to grayscale and normalized to the range $[-1, 1]$ before being fed into mPixelCNN++. During the detection stage, the ICA components are processed using the same per-band normalization strategy and mapped to $[-1, 1]$ before likelihood estimation. We also use the random flip and rotation data augmentation to stabilize the training. The model weights from the final epoch are used for subsequent experiments. The training process is conducted on an NVIDIA RTX A6000 GPU and takes approximately 31 h to complete. In addition to the default setting of $m = 2$, an alternative configuration with $m = 1$, which aligns with the original PixelCNN++, is employed. In the detection phase, all DL-based methods,

[6]https://github.com/zephyrhours/Hyperspectral-Anomaly-Detection-2S-GLRT

[7]https://github.com/xudongzhao461/Hyperspectral-Anomaly-Detection-by-Fractional-Fourier-Entropy/tree/master

[8]https://github.com/PuhongDuan/Hyperspectral-anomaly-detection-with-kernel-isolation-forest

[9]https://github.com/pei-xiang/GAED

[10]https://github.com/RSIDEA-WHU2020/Auto-AD

[11]https://github.com/jeline0110/GT-HAD/tree/main

[12]https://github.com/ChengXi-1217/DFAN-HAD

[13]https://github.com/ZhaoxuLi123/AETNet

[14]https://github.com/majitao-xd/BSDM-HAD

[15]https://github.com/JzXu123/TransGCF

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

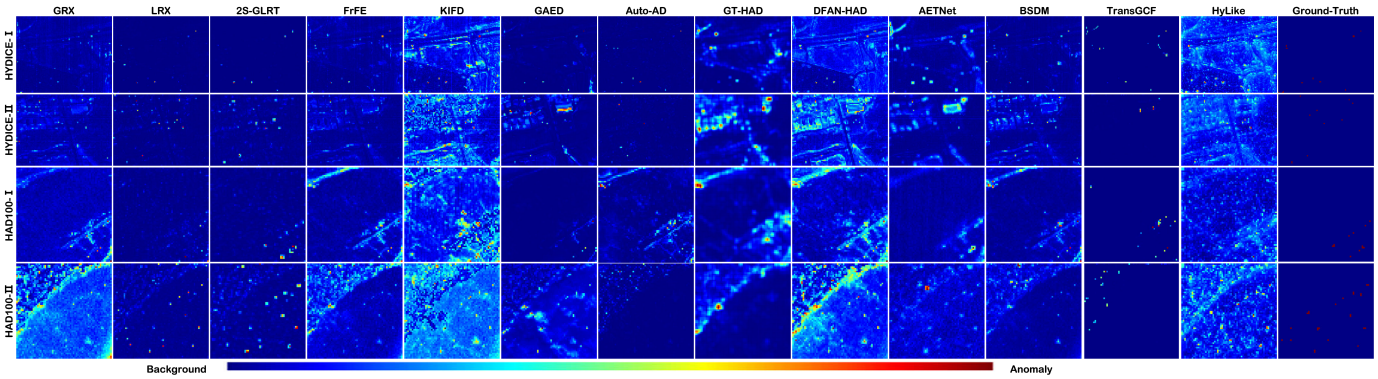


Fig. 9. Visual comparison of 13 methods on four HSIs. Methods are indicated along the top; images are listed vertically.

TABLE II

PERFORMANCE COMPARISON OF EIGHT AUC METRICS FOR 13 METHODS ON HYDICE-I AND HYDICE-II

Method	HYDICE-I								HYDICE-II							
	$AUC_{(D,F)}\uparrow$	$AUC_{ADP}\uparrow$	$AUC_{BDP}\uparrow$	$AUC_{JAD}\uparrow$	$AUC_{JBS}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{ADBS}\uparrow$	$AUC_{OADP}\uparrow$	$AUC_{(D,F)}\uparrow$	$AUC_{ADP}\uparrow$	$AUC_{BDP}\uparrow$	$AUC_{JAD}\uparrow$	$AUC_{JBS}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{ADBS}\uparrow$	$AUC_{OADP}\uparrow$
GRX	0.9849	0.2383	0.9658	1.2233	1.9507	6.9613	1.2041	2.1890	0.9274	0.1634	0.9435	1.0908	1.8708	2.8906	1.1069	2.0343
LRX	0.9919	0.1902	**0.9961**	1.1820	**1.9880**	49.1850	1.1863	2.1782	**0.9929**	0.3615	**0.9855**	1.3544	**1.9784**	**25.0111**	1.3471	**2.3399**
2S-GLRT	0.9813	0.2009	**0.9978**	1.1822	1.9791	**91.0993**	1.1986	2.1800	0.9709	0.2653	0.9783	1.2362	**1.9492**	**12.2358**	1.2436	2.2145
FrFE	0.9948	0.4210	0.9681	1.4157	1.9628	13.1781	1.3890	2.3838	0.9449	0.1509	0.9566	1.0958	1.9015	3.4761	1.1075	2.0524
KIFD	**0.9971**	**0.7429**	0.8681	**1.7400**	1.8652	5.6329	**1.6110**	**2.6082**	0.8933	**0.5652**	0.8262	**1.4585**	1.7195	3.2512	**1.3913**	2.2846
GAED	0.9845	0.1880	0.9886	1.1725	1.9731	16.4910	1.1766	2.1611	**0.9794**	0.3012	0.9555	1.2806	1.9349	6.7669	1.2567	2.2361
Auto-AD	0.9870	0.2259	0.9955	1.2129	**1.9825**	**49.7619**	1.2214	2.2084	0.9436	0.1520	**0.9920**	1.0955	1.9355	**18.9161**	1.1439	2.0875
GT-HAD	0.8964	0.1635	0.9523	1.0599	1.8487	3.4249	1.1157	2.0122	0.7287	0.1965	0.9006	0.9252	1.6293	1.9760	1.0970	1.8258
DFAN-HAD	**0.9961**	**0.5959**	0.8868	**1.5920**	1.8829	5.2631	**1.4827**	**2.4788**	0.9648	**0.5189**	0.8410	**1.4836**	1.8058	3.2639	**1.3599**	**2.3247**
AETNet	0.9917	0.5084	0.9535	1.5001	1.9452	10.9316	1.4619	2.4536	0.8873	0.1661	0.9422	1.0534	1.8296	2.8761	1.1083	1.9957
BSDM	0.9950	0.2857	0.9775	1.2808	1.9726	12.7248	1.2633	2.2583	0.9755	0.2899	0.9426	1.2654	1.9181	5.0496	1.2325	2.2080
TransGCF	0.9930	0.2099	**0.9980**	1.2029	**1.9910**	**107.0995**	1.2079	2.2009	0.9585	0.0085	**0.9977**	0.9671	**1.9562**	3.6728	1.0062	1.9647
HyLike	**0.9981**	**0.6531**	0.8342	**1.6512**	1.8322	3.9384	**1.4873**	**2.4853**	**0.9947**	**0.5237**	0.8393	**1.5185**	1.8341	3.2596	**1.3631**	**2.3578**

Note: The best, second-best, and third-best results are highlighted in bold red, bold blue, and bold black, respectively.

except for GAED, are implemented using PyTorch 1.13.1 and CUDA 11.6 on an NVIDIA GeForce RTX 4060 GPU. The remaining methods are executed in MATLAB R2022b on a system equipped with an Intel Core i9-13900H CPU @ 2.60 GHz and 16 GB of RAM.

C. Comparison of Performance

1) *Detection Maps:* Fig. 9 presents the colorized detection maps with all results min–max normalized to $[0, 1]$. HyLike achieves satisfactory anomaly enhancement with moderate FARs across the four HSIs. In contrast, the other methods fail to suppress the background and highlight the anomalies simultaneously. Specifically, GT-HAD and AETNet exhibit mosaic artifacts, especially on HYDICE-I and HYDICE-II. DFAN-HAD produces strong responses to textured regions on HAD100-I and HAD100-II, a tendency also observed in the results of KIFD. While GRX, FrFE, GAED, Auto-AD, BSDM, and TransGCF maintain relatively low FARs, their responses to anomalies are limited, as seen on HAD100-II. Window-based methods offer clean backgrounds but weak anomaly responses, notably on HYDICE-I. Overall, although HyLike produces certain background responses, the targets remain clearly distinguishable in the detection maps.
2) *3-D ROC Curve and Box–Whisker Plots:* The normalized 3-D ROC curves along with their corresponding three 2-D ROC curves of (PD, FAR), (PD, τ), and (FAR, τ) are depicted in Fig. 10. The 2-D ROC curve of (PD, FAR) for HyLike is closest to the upper left corner on HAD100-I and HAD100-II, indicating the best performance. On HYDICE-I, DFAN-HAD initially achieves a higher PD value than HyLike at low FAR levels. However, as the FAR increases, HyLike gradually surpasses DFAN-HAD. Due to the logarithmic scale on the horizontal axis, HyLike maintains a higher PD. A similar trend can be observed on HYDICE-II, where LRX achieves better performance only at very low FAR levels, while HyLike gradually outperforms LRX as the FAR increases. From the 2-D ROC of (PD, τ), it can be observed that our HyLike consistently remains highly competitive, positioned in close proximity to the upper right corner of the coordinate axis, demonstrating extremely high detection capability. Although our method exhibits high FARs on HYDICE-I and HYDICE-II, its FARs on HAD100-I and HAD100-II are lower than those of KIFD and GRX, respectively. In Fig. 11, the box–whisker plots evaluate different methods' background suppression and anomaly enhancement abilities. A flatter background box indicates stronger background suppression, while a larger margin between the background and anomaly boxes suggests a greater ability to enhance anomalies. On HYDICE-II and HAD100-II, HyLike exhibits the largest margin between background and anomaly boxes, indicating outstanding performance in anomaly enhancement. While this margin is not the largest on other HSIs, visually, HyLike achieves a reasonable

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

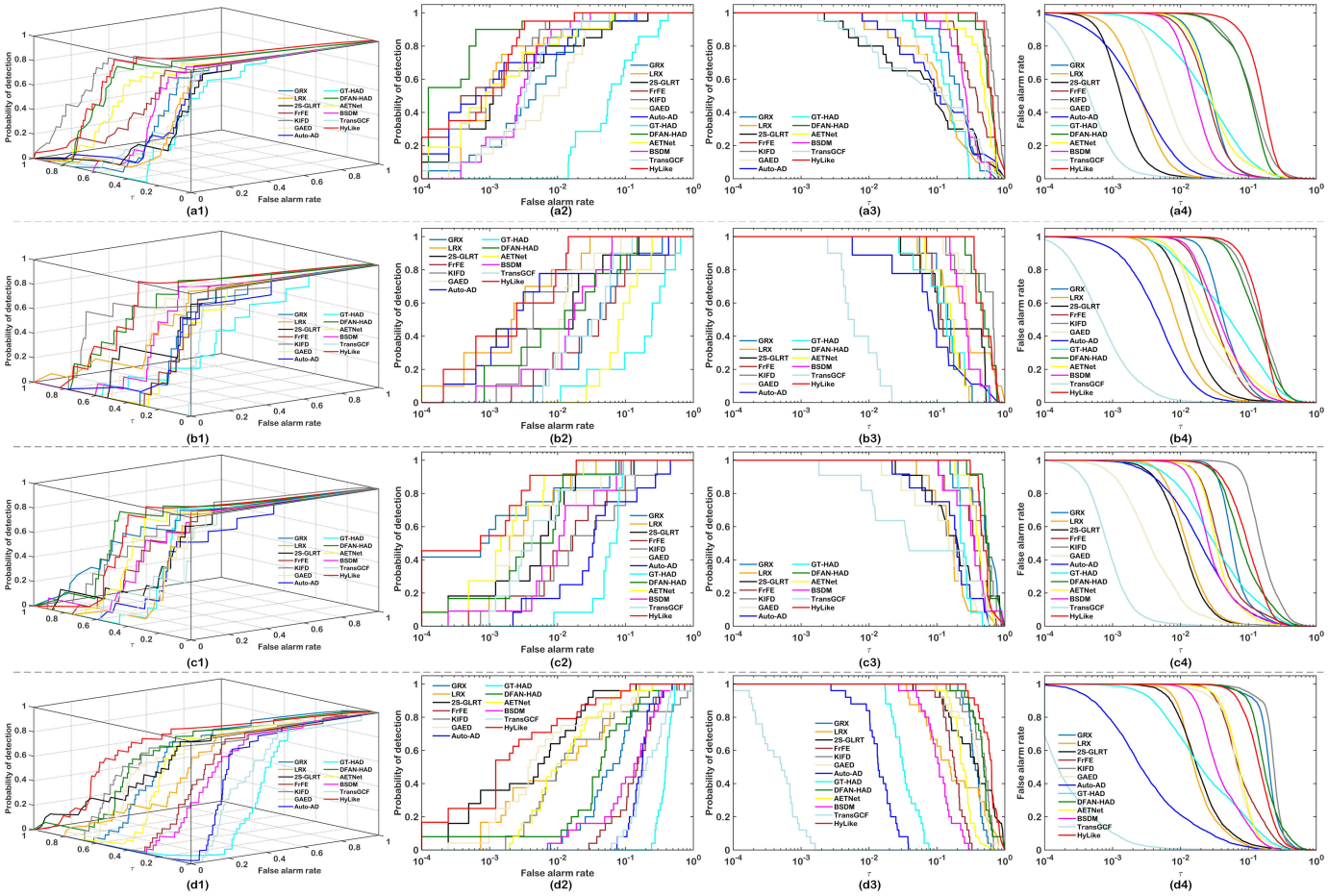


Fig. 10. Normalized 3-D ROC curves and corresponding 2-D ROC curves of different methods on four datasets. (a) HYDICE-I. (b) HYDICE-II. (c) HAD100-I. (d) HAD100-II. (a1)–(d1) 3-D ROC curves. (a2)–(d2) 2-D ROC curves of (PD, FAR). (a3)–(d3) 2-D ROC curves of (PD, τ). (a4)–(d4) 2-D ROC curves of (FAR, τ).

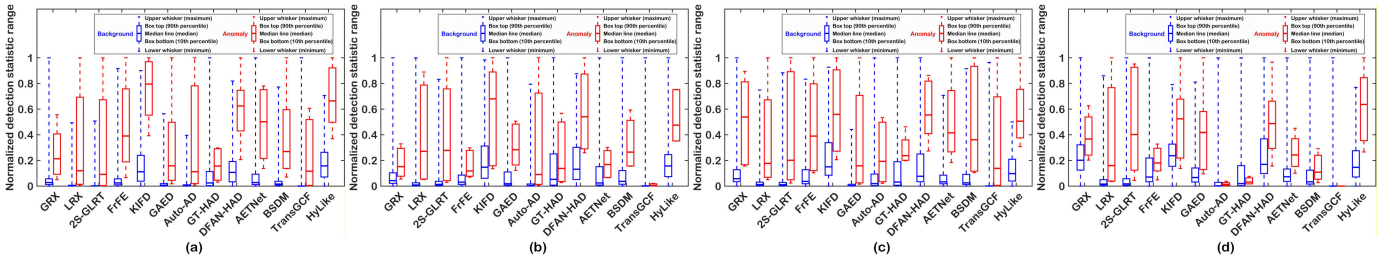


Fig. 11. Box–whisker plots of different methods on four HSIs. (a) HYDICE-I. (b) HYDICE-II. (c) HAD100-I. (d) HAD100-II.

compromise between background suppression and anomaly enhancement. On HYDICE-I and HAD100-I, HyLike exhibits similar detection statistics to those of competitive methods, particularly KIFD and AETNet. In addition, HyLike achieves lower maximum background responses and demonstrates overall superior separability on both datasets. Although box–whisker plots visually illustrate the background–anomaly separation of various algorithms, a representation confined to empirical $[10\%, 90\%]$ intervals cannot fully quantify background suppression and anomaly detection performance at continuous thresholds (τ) [75]. Consequently, eight comprehensive AUC metrics are further introduced for rigorous quantitative evaluation.

3) *Eight AUC Metrics:* The eight AUC values over four datasets are shown in Tables II and III. On the HYDICE-I dataset, KIFD achieves superior detection performance in $\mathrm{AUC_{ADP}}$ and $\mathrm{AUC_{JAD}}$, while HyLike still maintains competitive overall performance with an $\mathrm{AUC_{(D,F)}}$ of 0.9981. On the HYDICE-II dataset, LRX demonstrates effective background suppression capability, while DFAN-HAD exhibits notable detection performance. Meanwhile, HyLike achieves relatively robust overall performance. On the HAD100-I dataset, DFAN-HAD achieves strong detection performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PERFORMANCE COMPARISON OF EIGHT AUC METRICS FOR 13 METHODS ON HAD100-I AND HAD100-II

Method	HAD100-I								HAD100-II							
	$AUC_{(D,F)}\uparrow$	$AUC_{ADP}\uparrow$	$AUC_{BDP}\uparrow$	$AUC_{JAD}\uparrow$	$AUC_{JBS}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{ADBS}\uparrow$	$AUC_{OADP}\uparrow$	$AUC_{(D,F)}\uparrow$	$AUC_{ADP}\uparrow$	$AUC_{BDP}\uparrow$	$AUC_{JAD}\uparrow$	$AUC_{JBS}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{ADBS}\uparrow$	$AUC_{OADP}\uparrow$
GRX	0.9878	**0.5557**	0.9265	**1.5436**	1.9144	7.5637	**1.4823**	**2.4701**	0.9031	0.3825	0.7835	1.2856	1.6866	1.7666	1.1660	2.0691
LRX	**0.9931**	0.1986	0.9828	1.1917	**1.9758**	11.5246	1.1814	2.1745	0.9638	0.2857	**0.9740**	1.2495	**1.9378**	**10.9710**	1.2597	2.2235
2S-GLRT	0.9822	0.3089	**0.9834**	1.2911	1.9655	**18.5790**	1.2923	2.2744	**0.9857**	0.4534	0.9697	**1.4392**	**1.9554**	**14.9481**	**1.4231**	**2.4088**
FrFE	0.9671	0.3729	0.9383	1.3400	1.9054	6.0409	1.3112	2.2783	0.8212	0.1881	0.8963	1.0093	1.7175	1.8140	1.0844	1.9056
KIFD	0.9377	**0.5197**	0.8147	1.4575	1.7524	2.8046	1.3344	2.2722	0.8663	**0.4749**	0.7579	1.3412	1.6242	1.9617	1.2328	2.0991
GAED	0.9843	0.2102	**0.9921**	1.1944	**1.9763**	**26.4672**	1.2022	2.1865	0.9691	0.3920	0.9179	1.3611	1.8870	**4.7728**	**1.3098**	**2.2789**
Auto-AD	0.9079	0.2211	0.9595	1.1289	1.8674	5.4605	1.1806	2.0884	0.8149	0.0164	**0.9883**	0.8313	1.8032	1.3969	1.0047	1.8196
GT-HAD	0.9381	0.2614	0.9308	1.1996	1.8690	3.7787	1.1922	2.1304	0.6086	0.0372	0.9430	0.6458	1.5516	0.6522	0.9802	1.5888
DFAN-HAD	0.9882	**0.5797**	0.8865	**1.5679**	1.8747	5.1083	**1.4662**	**2.4544**	0.9211	**0.4978**	0.7947	**1.4189**	1.7159	2.4252	1.2925	2.2137
AETNet	**0.9958**	0.4206	0.9539	1.4164	1.9497	9.1214	1.3745	2.3703	**0.9725**	0.2594	0.9172	1.2319	**1.8897**	3.1308	1.1765	2.1491
BSDM	0.9767	0.3733	0.9538	1.3500	1.9305	8.0719	1.3270	2.3038	0.8397	0.1235	0.9440	0.9632	1.7837	2.2025	1.0674	1.9071
TransGCF	0.9825	0.2009	**0.9970**	1.1834	**1.9794**	**66.1437**	1.1978	2.1803	0.7390	0.0006	**0.9973**	0.7396	1.7364	0.2167	0.9979	1.7369
HyLike	**0.9973**	0.5081	0.8860	**1.5053**	1.8833	4.4583	**1.3941**	**2.3914**	**0.9867**	**0.6071**	0.8352	**1.5938**	1.8219	3.6844	**1.4423**	**2.4290**

Note: The best, second-best, and third-best results are highlighted in bold red, bold blue, and bold black, respectively.

TABLE IV

PERFORMANCE COMPARISON OF MAUC METRICS FOR 13 METHODS AVERAGED OVER FOUR DATASETS

Method	$mAUC_{(D,F)}\uparrow$	$mAUC_{ADP}\uparrow$	$mAUC_{BDP}\uparrow$	$mAUC_{JAD}\uparrow$	$mAUC_{JBS}\uparrow$	$mAUC_{SNPR}\uparrow$	$mAUC_{ADBS}\uparrow$	$mAUC_{OADP}\uparrow$
GRX	0.9508	0.3350	0.9048	1.2858	1.8556	4.7956	1.2398	2.1906
LRX	**0.9854**	0.2590	**0.9846**	1.2444	**1.9700**	**24.1729**	1.2436	2.2290
2S-GLRT	**0.9800**	0.3071	0.9823	1.2872	**1.9623**	**34.2156**	1.2894	2.2694
FrFE	0.9320	0.2832	0.9398	1.2152	1.8718	6.1272	1.2230	2.1550
KIFD	0.9236	**0.5757**	0.8167	**1.4993**	1.7403	3.4126	**1.3924**	**2.3160**
GAED	0.9793	0.2728	0.9635	1.2521	**1.9428**	13.6245	1.2363	2.2156
Auto-AD	0.9133	0.1538	**0.9838**	1.0672	1.8971	18.8838	1.1376	2.0510
GT-HAD	0.7930	0.1646	0.9317	0.9576	1.7246	2.4580	1.0963	1.8893
DFAN-HAD	0.9675	**0.5481**	0.8523	**1.5156**	1.8198	4.0151	**1.4003**	**2.3679**
AETNet	0.9618	0.3386	0.9417	1.3005	1.9035	6.5150	1.2803	2.2421
BSDM	0.9467	0.2681	0.9545	1.2148	1.9012	7.0122	1.2226	2.1693
TransGCF	0.9183	0.1050	**0.9975**	1.0232	1.9158	**44.2832**	1.1025	2.0207
HyLike	**0.9942**	**0.5730**	0.8487	**1.5672**	1.8429	3.8352	**1.4217**	**2.4159**

Note: The best, second-best, and third-best results are highlighted in bold red, bold blue, and bold black, respectively.

TABLE V

EXECUTION TIME (IN SECONDS) OF 13 METHODS ON FOUR HSIS

Dataset\Method	GRX	LRX	2S-GLRT	FrFE	KIFD	GAED	Auto-AD	GT-HAD	DFAN-HAD	AETNet	BSDM	TransGCF	HyLike
HYDICE-I	**0.0316**	**2.6582**	10.1309	5.1985	11.0480	31.7255	30.9372	22.0970	11.2489	**0.1644**	10.6386	70.1210	16.5507
HYDICE-II	**0.0192**	**1.6114**	6.0126	3.2439	9.9072	8.7558	19.2511	11.3937	4.9173	**0.1338**	9.9146	48.7600	11.0787
HAD100-I	**0.0305**	5.1853	14.4928	14.7158	5.3254	12.0040	**4.3744**	14.3672	5.3673	**0.1140**	10.4229	51.1149	13.1047
HAD100-II	**0.0276**	5.2657	14.6189	14.8154	7.8740	22.2664	18.2447	18.5311	**4.0193**	**0.1108**	10.8034	53.9382	13.7559
Average	**0.0272**	**3.6801**	11.3138	9.4934	8.5386	18.6879	18.2018	16.5973	6.3882	**0.1308**	10.4449	55.9835	13.6225

Note: The lowest, second-lowest, and third-lowest time consumptions are highlighted by bold red, bold blue, and bold black fonts, respectively.

in AUC_{ADP} and AUC_{JAD}, whereas TransGCF shows superior background suppression performance. On the HAD100-II dataset, HyLike achieves the best overall detection performance, ranking first in five evaluation metrics, followed by 2S-GLRT, which demonstrates superior background suppression capability. Overall, HyLike ranks within the top three in 19 out of 32 AUC evaluations across four datasets, demonstrating consistently competitive detection performance under diverse metrics. Furthermore, the mean AUC (mAUC) metrics are summarized in Table IV. HyLike ranks first in $mAUC_{(D,F)}$, $mAUC_{JAD}$, $mAUC_{ADBS}$, and $mAUC_{OADP}$, and second in $mAUC_{ADP}$. Notably, the background suppression ability of HyLike is limited. This phenomenon may be attributed to potential degradation in prediction accuracy due to the shift displacement (see more detailed explanations in Section IV-E, item 3) and the fusion strategy, which will be further investigated in our future work.

4) *Time Consumption:* To comprehensively evaluate the performance, the computational efficiency is reported in Table V with the last row averaging the time consumption across all HSIs. Note that, in Table V, all DL methods except AETNet and HyLike report the total time including both training and detection (inference) stages. In contrast, AETNet and HyLike report only the complete detection-stage time, including all necessary processing steps from HSI input to anomaly detection output since these methods do not require retraining after the initial pretraining stage. Although HyLike exhibits moderate time consumption, it ranks fourth among the DL-based methods. In addition, HyLike utilizes Auto-ICA to enable parameter-free detection, albeit at the cost of repeating the detection process ten times to determine the optimal dimensionality (see Algorithm 2). With its previously demonstrated superior detection performance, we demonstrate that HyLike achieves the optimal overall performance among all compared methods.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

NLL ON THE TEST SET FOR MODELS TRAINED ON GRAYSCALE CIFAR-10 WITH DIFFERENT SHIFT DISPLACEMENT SETTINGS (m)

Model	Bits per pixel
PixelCNN++($m = 1$)	4.73
mPixelCNN++($m = 2$)	5.89

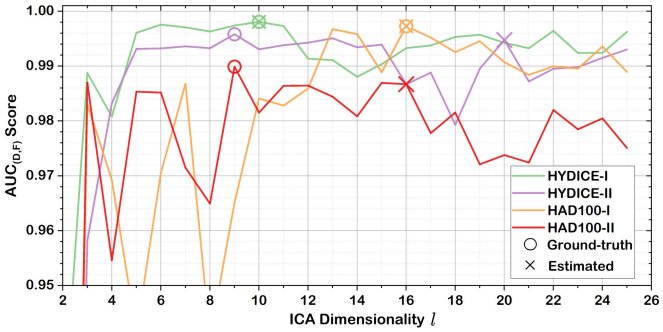


Fig. 12. $AUC_{(D,F)}$ versus the number of ICs. The colored markers $\times$ and $\circ$ indicate the AUC scores at the estimated optimal number of ICs (given by our Auto-ICA) and the ground-truth optimal number (obtained via exhaustive search), respectively.

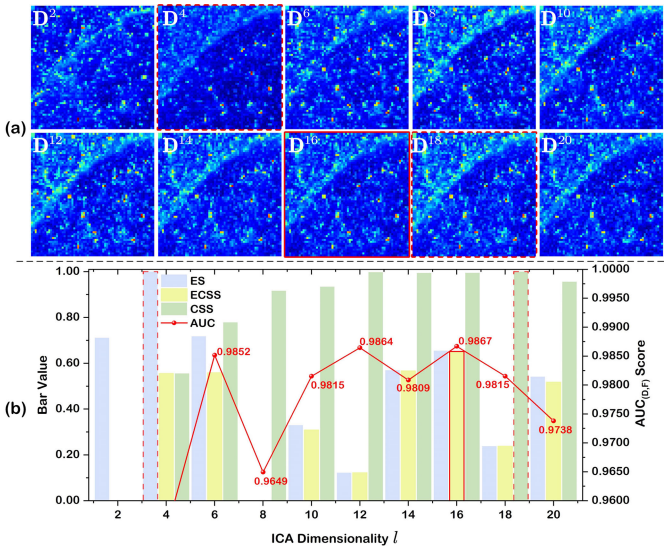


Fig. 13. Bar chart illustrating the impact of ECSS on dimensionality selection. (a) Detection maps at various ICA dimensionality. ECSS-selected map marked by solid red box. ES and CSS are selected by red dashed boxes. (b) Corresponding ES, CSS, ECSS, and $AUC_{(D,F)}$ values calculated from $\mathbf{D}^l$.

D. Generative Ability

The NLL on the test set of the original PixelCNN++ ($m = 1$) and our mPixelCNN++ ($m = 2$) is reported in Table VI, where a smaller NLL value implies that the model assigns higher probabilities to the observed data, indicating a better fit to the true data distribution. Here, bits per pixel are computed as the average NLL per pixel over the test set. While the predictive performance of mPixelCNN++ is naturally lower due to the lack of adjacent pixel information, corresponding to the decreased R^2 in Fig. 7, it is still capable of modeling pixel-wise dependencies using nonadjacent upper left context. Importantly, our goal is to learn general pixel-wise likelihood priors rather than to generate images from a specific distribution. From this perspective, the reported NLL remains sufficient for the HAD task.

E. Ablation Study

This section presents a comprehensive evaluation of each module of HyLike. We first assess the effectiveness of the Auto-ICA module by comparing the $AUC_{(D,F)}$ achieved at the automatically determined dimensionality (L^*) with the best $AUC_{(D,F)}$ obtained by manual selection of the dimensionality (L). Subsequently, we evaluate the effectiveness of the ICA transformation, mPixelCNN++, and LRX-WFS under the fixed dimensionality L^*. For clarity, we define **Case1** as the configuration where detection is conducted in the original spectral space (i.e., without ICA transformation), **Case2** as the setting in which mPixelCNN++ is replaced with the original PixelCNN++ [i.e., m is set to 1 in (6) and (7)], and **Case3** as the setting where all weights in (17) are set to 1 (i.e., without LRX-WFS). The quantitative ablation results are depicted in Table VII.

1) *Effectiveness of Auto-ICA*: To quantify the difference between the AUC achieved at L^* and that at L, we traverse the ICA dimensionality l from 2 to 25 and collect the $AUC_{(D,F)}$ under each dimensionality across four HSIs. The result is shown in Fig. 12. Notably, the difference between the best $AUC_{(D,F)}$ and that obtained using the dimensionality determined by our Auto-ICA is small, especially for HAD100-I and HYDICE-I, where they are equal. This demonstrates the effectiveness of Auto-ICA. Although some HSIs achieve their best $AUC_{(D,F)}$ beyond HyLike's search range, the comparison experiments demonstrate that traversing dimensionality from 2 to 20 in steps of 2 is sufficient.
To further investigate the influence of each part of ECSS in Auto-ICA, the visualization of entropy score (ES) and CSS, calculated from the detection map $\mathbf{D}^l$ of HAD100-II, is shown in Fig. 13. The detection maps exhibiting acceptable detection performance often display high global similarity in visual content, as observed in $\mathbf{D}^{16}$ and $\mathbf{D}^{18}$. This observation motivates the introduction of CSS as a complementary metric to ES. Although $\mathbf{D}^4$ and $\mathbf{D}^{18}$ have the highest ES and CSS, respectively, their AUC values are lower than the optimal AUC, which is correctly identified by ECSS, confirming the effectiveness of Auto-ICA. To ensure the controlled experimental conditions, we fix the dimensionality L^* determined by our Auto-ICA in the ICA transformation throughout the subsequent ablation experiment.
2) *Effectiveness of ICA*: **Case1** in Table VII reports the $AUC_{(D,F)}$ values obtained in the original spectral space, i.e., without ICA transformation. None of these AUC values for four HSIs surpass those of HyLike. Moreover, the total number of processing bands is fixed at 110 in our HyLike, whereas direct detection in the original spectral space must handle a much larger number of bands, resulting in substantially higher computational costs without corresponding performance gains. This highlights the efficiency and effectiveness of using ICA

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

$AUC_{(D,F)}$ VALUES (EXECUTION TIME IN SECONDS) FOR DIFFERENT CASES ON FOUR HYPERSPECTRAL DATASETS

Dataset&Processing\Case	Case1	Case2	Case3	HyLike
ICA		✓	✓	✓
mPixelCNN++	✓		✓	✓
LRX-WFS	✓	✓		✓
HYDICE-I	0.9901 (16.9591)	0.9768 (1.3177)	0.9964 (**1.1985**)	**0.9981** (1.4218)
HYDICE-II	0.9481 (11.2195)	0.9803 (1.6183)	0.9897 (**1.4520**)	**0.9947** (1.7994)
HAD100-I	0.9689 (18.3194)	0.9925 (1.7771)	0.9670 (**1.4227**)	**0.9973** (2.0644)
HAD100-II	0.9768 (16.6932)	0.9148 (1.8428)	0.8846 (**1.4594**)	**0.9867** (2.0216)
Average	0.9710 (15.7978)	0.9661 (1.6390)	0.9595 (**1.3832**)	**0.9942** (1.8268)

Note: bold black fonts highlight the best performance.

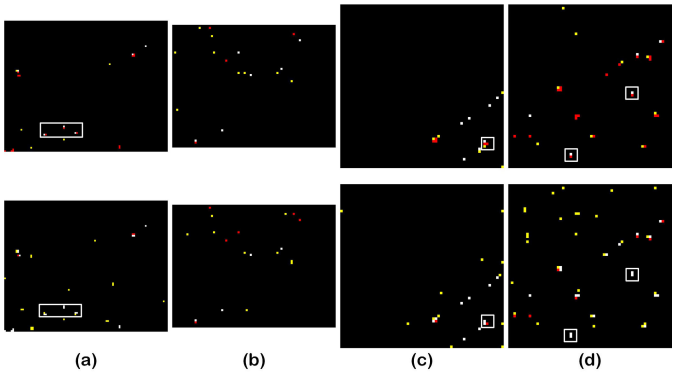


Fig. 14. Comparison of detection results across four HSIs using the original PixelCNN++ (first row) and mPixelCNN++ (second row), with correct, missed, and false detections labeled in white, red, and yellow, respectively. Improvements are highlighted with white boxes. (a) HYDICE-I. (b) HYDICE-II. (c) HAD100-I. (d) HAD100-II.

in our HyLike framework. The last column of Table VII shows that, given the ICA dimensionality prior, HyLike achieves an average processing time of 1.8 s, which is substantially faster than performing detection in the original spectral space.

3) *Effectiveness of mPixelCNN++*: The $AUC_{(D,F)}$ values of HyLike with the original PixelCNN++ ($m = 1$) are summarized in Table VII (**Case2**). It can be observed that mPixelCNN++ consistently achieves higher AUC values than the original PixelCNN++ across all four datasets. To better illustrate this phenomenon, the detection maps are binarized using (14), where correct detections, missed detections, and false detections are marked in white, red, and yellow, respectively. As shown in Fig. 14, although mPixelCNN++ produces more false alarms (yellow pixels) than the original PixelCNN++, it can effectively detect most anomalies spanning multiple pixels (see the white boxes in Fig. 14). This phenomenon arises because, with $m = 1$, background pixels are easily predicted from neighbors, and the same holds for small spatially contiguous anomalies. As a result, the original PixelCNN++ tends to produce lower NLL values for both background and anomaly regions, thereby reducing false alarms while simultaneously increasing missed detections, which is a more critical issue than high FARs. Overall, both the quantitative and visual results confirm the effectiveness of the proposed mPixelCNN++.

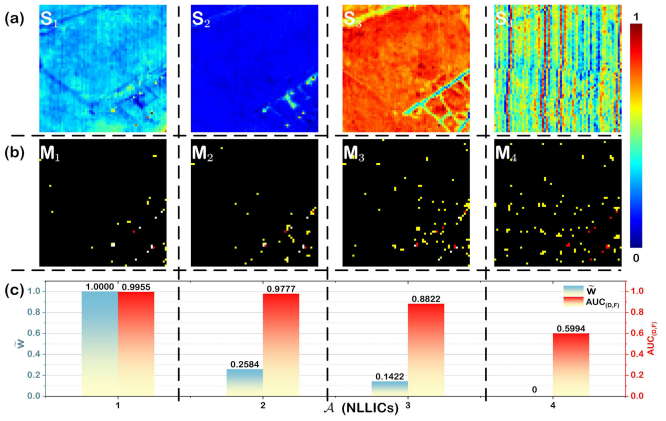


Fig. 15. Visualization of ICs and NLLICs of HAD100-I produced by our HyLike. (a) Subset of ICs. (b) Binarized NLLICs corresponding to (a), where missed anomalous pixels, false detections, and correctly identified pixels are shown in red, yellow, and white, respectively. (c) $AUC_{(D,F)}$ values and weights derived from the corresponding NLLICs.

4) *Effectiveness of LRX-WFS*: The $AUC_{(D,F)}$ values obtained by assigning equal weights to all NLLICs (Table VII, **Case3**) indicate that simple summation is a suboptimal strategy. To further illustrate this issue, a subset of ICs $\mathcal{S}$ produced by HyLike on HAD100-I is manually selected. The corresponding binary detection maps $\mathcal{M}$, derived for NLLICs $\mathcal{A}$, weights $\tilde{\mathbf{w}}$, and AUC scores calculated from $\mathcal{A}$ are shown in Fig. 15. As seen in Fig. 15(a), the IC corresponding to the noise component ($\mathbf{S}_4$) is effectively suppressed by our LRX-WFS through the assignment of zero weight. Fig. 15(b) confirms that different ICs contribute unequally, and appropriate fusion may enhance detection (see $\mathbf{M}_2$ and $\mathbf{M}_3$). Fig. 15(c) further shows that the assigned weights are approximately proportional to the AUC values, highlighting the positive contributions of each NLLIC to anomaly detection. In a word, our LRX-WFS adaptively emphasizes the NLLICs that significantly contribute to anomaly detection, resulting in better detection performance than simple summation. The sensitivity of the LRX-WFS window parameters is analyzed by testing several window combinations. The mean $AUC_{(D,F)}$ and the mean execution time averaged over the four hyperspectral datasets are reported in Table VIII. As shown in Table VIII, the configurations (3, 5) and (3, 7) achieve almost identical $mAUC_{(D,F)}$ values, while the larger outer window leads to increased computational cost. Therefore, the configuration (3, 5) is adopted in this work as it provides a favorable tradeoff between detection accuracy and computational efficiency.

F. Performance Analysis of Target Size

This section analyzes the impact of target size on the detection performance of our HyLike. Although using simulated datasets to validate algorithm performance cannot fully reflect what a real situation is [75], we still adopt a linear simulation approach to achieve a simplified, controllable, and identical background distribution, to explore the relationship between the displacement parameter and anomaly size. To this end, we

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

SENSITIVITY ANALYSIS OF LRX-WFS WINDOW PARAMETERS. MEAN $AUC_{(D,F)}$ AND MEAN TIME (IN SECONDS) AVERAGED OVER THE FOUR DATASETS

window parameters	mean $AUC_{(D,F)}$	mean time
(3,5)	0.9942	1.8268
(3,7)	0.9946	1.9988
(3,9)	0.9932	2.1087
(5,7)	0.9928	1.9809
(5,9)	0.9920	2.0720
(7,9)	0.9884	2.0792

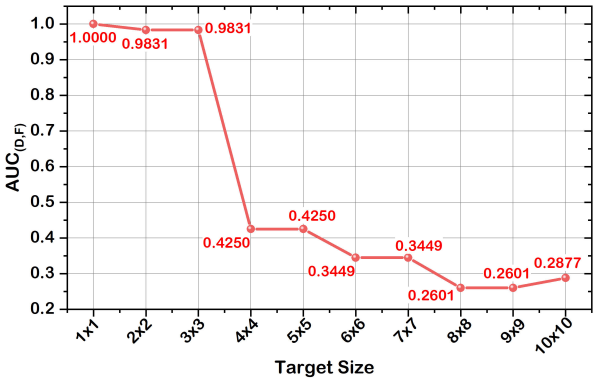


Fig. 16. $AUC_{(D,F)}$ values of HyLike for different target sizes on ten synthetic HSIs.

simulate anomalies of different sizes on the same background HSI using a linear mixing model. Specifically, anomalous pixels are simulated by implanting a target spectral signature into background pixels using an LMM: $\mathbf{x}_i^t = f_i\mathbf{t} + (1- f_i)\mathbf{b}_i$, where $\mathbf{x}_i^t$ denotes the simulated spectrum of pixel i, $\mathbf{b}_i$ is the original spectrum of the ith background pixel, $\mathbf{t}$ is the given anomaly target spectrum, and $f_i \in (0, 1]$ is the mixing ratio between the target and the background. We implant anomalous targets arranged in a 3×3 spatial grid pattern onto the widely used Salinas Valley dataset, which contains only background pixels and has been commonly employed as a pure-background benchmark for anomaly simulation [73], with a fixed mixing ratio $f_i = 0.5$. The target size varies from 1×1 to 10×10. The $AUC_{(D,F)}$ scores achieved by HyLike are illustrated in Fig. 16. It is clear that the AUC score is severely affected by the target size; the larger size produces worse detection. Because the shift displacement m determines the receptive field of HyLike, in our mPixelCNN++, m is set to 2, meaning that the adjacent upper, left, and upper left pixels are excluded in the prediction of the current pixel. Thus, target size beyond m will result in the participation of the anomaly in the prediction, causing lower NLL for anomalous pixels. Therefore, HyLike faces the challenge of detecting these large anomaly targets. However, the spatial resolution of most currently flying satellites, such as ZY1E, EnMap, and GF5-AHSI, is relatively coarse at around 30 m, causing anomalies to occupy only one or two pixels in the image typically. In such cases, HyLike demonstrates superior performance in HAD tasks.

G. Extension With EAS and ISSMax

To further evaluate the effectiveness of the proposed HyLike, we integrate HyLike with the effective anomaly space (EAS) [76] and the iterative spectral–spatial maximal map (ISSMax) [75]. Specifically, two hybrid detection frameworks are implemented: 1) performing anomaly detection directly in the EAS domain (EAS-HyLike) and 2) performing anomaly detection jointly in the EAS domain and the iterative refinement process (EAS-ISSMax-HyLike). In these implementations, the ICA transformation stage in HyLike is removed, and the EAS-transformed representation is directly fed into the network as input. The dimensionality parameter j in the EAS transformation is empirically selected from the range of 2–25 to achieve optimal performance. In addition, HyLike is replaced with GRX to construct two comparative methods, namely, EAS-GRX and EAS-ISSMax-GRX. The corresponding experimental results are reported in Table IX. It can be observed that, after incorporating EAS and EAS-ISSMax, the performance of GRX is improved on most datasets, except for HAD100-II. In particular, EAS-ISSMax-GRX achieves better results than HyLike on the HAD100-I dataset. On the HYDICE-I dataset, EAS-ISSMax-HyLike outperforms the original HyLike in terms of AUC_{ADP}, AUC_{BDP}, and AUC_{OADP}, indicating the effectiveness of both EAS and ISSMax. In contrast, on HYDICE-II and HAD100-II, HyLike

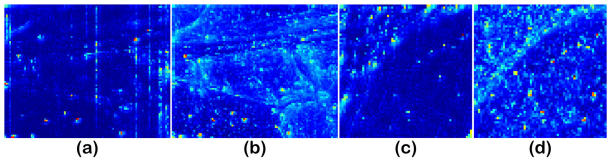


Fig. 17. Comparison between EAS-HyLike and HyLike on HYDICE-I and HAD100-II. (a) and (c) EAS-HyLike. (b) and (d) HyLike.

TABLE IX

PERFORMANCE OF GRX AND HYLIKE INTEGRATED WITH EAS AND ISSMAX ON FOUR DATASETS

Dataset	Methods	$AUC_{(D,F)}$ ↑	AUC_{ADP} ↑	AUC_{BDP} ↑	AUC_{OADP} ↑
HYDICE-I	GRX	0.9849	0.2383	**0.9658**	2.1890
	EAS-GRX	0.9955	0.3137	**0.9722**	2.2814
	EAS-ISSMax-GRX	0.9953	0.3142	**0.9721**	2.2817
	HyLike	**0.9981**	**0.6531**	0.8342	**2.4853**
	EAS-HyLike	**0.9959**	**0.6828**	0.9371	**2.6158**
	EAS-ISSMax-HyLike	**0.9960**	**0.7097**	0.9236	**2.6293**
HYDICE-II	GRX	0.9274	0.1634	**0.9435**	2.0343
	EAS-GRX	0.9526	0.3105	**0.9453**	2.2084
	EAS-ISSMax-GRX	0.9516	0.3121	**0.9444**	2.2082
	HyLike	**0.9947**	**0.5237**	0.8393	**2.3578**
	EAS-HyLike	**0.9694**	**0.4589**	0.8979	**2.3261**
	EAS-ISSMax-HyLike	**0.9701**	**0.4751**	0.8938	**2.3390**
HAD100-I	GRX	0.9878	**0.5557**	0.9265	**2.4701**
	EAS-GRX	**0.9932**	**0.8466**	**0.9336**	**2.7734**
	EAS-ISSMax-GRX	**0.9932**	**0.8473**	0.9334	**2.7739**
	HyLike	**0.9973**	0.5081	0.8860	2.3914
	EAS-HyLike	0.9897	0.3845	**0.9908**	2.3651
	EAS-ISSMax-HyLike	**0.9897**	0.3848	**0.9908**	2.3653
HAD100-II	GRX	**0.9031**	**0.3825**	0.7835	**2.0691**
	EAS-GRX	**0.8680**	0.2545	**0.9114**	2.0339
	EAS-ISSMax-GRX	0.8678	**0.2553**	0.9112	**2.0342**
	HyLike	**0.9867**	**0.6071**	0.8352	**2.4290**
	EAS-HyLike	0.8498	0.2250	**0.9191**	1.9939
	EAS-ISSMax-HyLike	0.8546	0.2303	**0.9209**	2.0057

Note: The best, second-best, and third-best results are highlighted in bold red, bold blue, and bold black, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

with EAS exhibits performance degradation in certain metrics while still achieving improved background suppression. This behavior may be attributed to the greater detection difficulty associated with these datasets (see Fig. 17). Furthermore, across all datasets, incorporating EAS consistently improves AUC_{BDP} for all methods compared with their original versions, suggesting that EAS is effective in reducing background interference. These results indicate that EAS provides a beneficial representation for background suppression, suggesting that further improvements to HyLike may be achieved by incorporating sparsity cardinality in the IC space to suppress background signals and enhance anomaly detection.

V. Conclusion

In this article, we propose HyLike, a novel HAD method that transfers the natural image prior to HAD by modeling pixel dependencies with pixel-wise likelihood modeling. Once trained on natural images, HyLike can detect anomalies in any HSI based on NLL without fine-tuning. In the prior learning phase, our proposed mPixelCNN++ effectively mitigates the unreliable low NLL caused by spatially contiguous anomalies, thereby achieving improved detection performance. In the detection phase, ICA transformation in HyLike can bridge the modality gap while lowering computational cost. With the proposed LRX-WFS, HyLike can emphasize and fuse key NLLICs to further boost detection capabilities. Furthermore, the proposed Auto-ICA renders HyLike a parameter-free detection method. Extensive experimental results demonstrate the effectiveness and superiority of HyLike. Future work will further improve HyLike from the perspective of background suppression. First, we will explore more anomaly-oriented representations in the ICA feature space before likelihood estimation. Sparse anomaly-relevant component enhancement inspired by EAS can be investigated to suppress background-dominated components before NLLICs generation. Second, more adaptive likelihood fusion strategies will be developed to dynamically emphasize reliable NLLICs with compact and high-contrast anomaly responses while reducing the influence of unreliable NLLICs. Third, more stable conditional probability modeling frameworks will be investigated to improve the robustness of pixel-wise likelihood estimation, thereby reducing undesired high NLL responses in background regions caused by missing or unreliable neighborhood information.

References

[1] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.

[2] N. Raza Shah et al., "Hyperspectral anomaly detection: A performance comparison of existing techniques," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 2078–2125, Dec. 2022.

[3] D. Haboudane, J. R. Miller, E. Pattey, P. J. Zarco-Tejada, and I. B. Strachan, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, Apr. 2004.

[4] S. Peyghambari and Y. Zhang, "Hyperspectral remote sensing in lithological mapping, mineral exploration, and environmental geology: An updated review," *J. Appl. Remote Sens.*, vol. 15, no. 3, Jul. 2021, Art. no. 031501.

[5] M. Shimoni, R. Haelterman, and C. Perneel, "Hyperspectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[6] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS^{3}LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[7] X. Sun, L. Zhuang, L. Gao, H. Gao, X. Sun, and B. Zhang, "Information entropy estimation based on point-set topology for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5523415.

[8] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[9] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.

[10] L. Zhuang and M. K. Ng, "FastHyMix: Fast and parameter-free hyperspectral image mixed noise removal," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4702–4716, Aug. 2023.

[11] L. Zhuang, M. K. Ng, L. Gao, J. Michalski, and Z. Wang, "Eigenimage2Eigenimage (E2E): A self-supervised deep learning network for hyperspectral image denoising," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16262–16276, Nov. 2024.

[12] L. Zhuang, M. K. Ng, L. Gao, and Z. Wang, "Eigen-CNN: Eigenimages plus eigennoise level maps guided network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512018.

[13] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[14] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[15] A. Schaum, "Joint subspace detection of hyperspectral targets," in *Proc. IEEE Aerosp. Conf.*, Mar. 2004, pp. 1818–1824.

[16] M. J. Carlotto, "A cluster-based approach for detecting man-made objects and changes in imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 374–387, Feb. 2005.

[17] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[18] R. Tao, X. Zhao, W. Li, H.-C. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[19] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, and H. Li, "Multipixel anomaly detection with unknown patterns for hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5557–5567, Oct. 2022.

[20] C.-I. Chang, "Target-to-anomaly conversion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540428.

[21] C.-I. Chang, C.-Y. Lin, P.-C. Chung, and P. F. Hu, "Iterative spectral–spatial hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504330.

[22] C.-I. Chang, C.-Y. Lin, P.-C. Chung, and P. Fuming Hu, "Cascaded spectral–spatial hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5504022.

[23] L. Zhuang, L. Gao, B. Zhang, X. Fu, and J. M. Bioucas-Dias, "Hyperspectral image denoising and anomaly detection based on low-rank and sparse representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500117.

[24] J. Peng et al., "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[25] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[26] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533417.

[27] L. Zhuang, X. Fu, M. K. Ng, and J. M. Bioucas-Dias, "Hyperspectral image denoising based on global and nonlocal low-rank factorizations," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10438–10454, Dec. 2021.

[28] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation,"

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

IEEE Trans. Geosci. Remote Sens., vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[29] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[30] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.

[31] Q. Ling, Y. Guo, Z. Lin, and W. An, "A constrained sparse representation model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2358–2371, Apr. 2019.

[32] H. Su, Z. Wu, Q. Du, and P. Du, "Hyperspectral anomaly detection using collaborative representation with outlier removal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 5029–5038, Dec. 2018.

[33] Y. Qu et al., "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[34] M. Feng, Y. Yang, X. Shao, and Q. Shu, "Hyperspectral anomaly detection based on multicomplementary prior-guided tensor decomposition," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–17, 2025.

[35] K. Yu et al., "Efficient low-rank representation for hyperspectral anomaly detection via pixel segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5509813.

[36] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[37] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[38] S. Sun, J. Liu, Z. Zhang, and W. Li, "Hyperspectral anomaly detection based on adaptive low-rank transformed tensor," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9787–9799, Jul. 2024.

[39] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[40] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[41] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[42] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022.

[43] D. Wang, L. Gao, Y. Qu, X. Sun, and W. Liao, "Frequency-to-spectrum mapping GAN for semisupervised hyperspectral anomaly detection," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1258–1273, Dec. 2023, doi: 10.1049/cit2.12154.

[44] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.

[45] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. J. Plaza, "PDBSNet: Pixel-shuffle downsampling blind-spot reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511914.

[46] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. Plaza, "BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531916.

[47] D. Wang, L. Zhuang, L. Gao, X. Sun, X. Zhao, and A. Plaza, "Sliding dual-window-inspired reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504115.

[48] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818.

[49] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3631–3645, Feb. 2025.

[50] X. Cheng et al., "Deep feature aggregation network for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.

[51] J. Ma, W. Xie, Y. Shi, X. Xiang, Y. Li, and L. Fang, "BSDM: Background suppression diffusion model for hyperspectral anomaly detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 1, pp. 190–204, Jan. 2026.

[52] J. Xu et al., "TransGCF: A unified spatial–spectral–frequency framework for robust hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 6875–6891, 2026.

[53] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021.

[54] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Deep low-rank prior for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527017.

[55] S. Lin, X. Cheng, Y. Zeng, Y. Huo, M. Zhang, and H. Wang, "Low-rank and sparse representation inspired interpretable network for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.

[56] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[57] W. Rao, Y. Qu, L. Gao, X. Sun, Y. Wu, and B. Zhang, "Transferable network with Siamese architecture for anomaly detection in hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102669.

[58] Z. Li, Y. Wang, C. Xiao, Q. Ling, Z. Lin, and W. An, "You only train once: Learning a general anomaly enhancement network with random masks for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506718.

[59] J. Li, X. Wang, H. Zhao, and Y. Zhong, "Learning a cross-modality anomaly detector for remote sensing imagery," *IEEE Trans. Image Process.*, vol. 33, pp. 6607–6621, 2024.

[60] W. Chen, X. Zhi, S. Jiang, Y. Huang, Q. Han, and W. Zhang, "DWSDiff: Dual-window spectral diffusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504617.

[61] E. P. Simoncelli and B. A. Olshausen, "Natural image statistics and neural representation," *Annu. Rev. Neurosci.*, vol. 24, no. 1, pp. 1193–1216, Mar. 2001.

[62] T. Salimans, A. Karpathy, X. Chen, and D. P. Kingma, "PixelCNN++: Improving the PixelCNN with discretized logistic mixture likelihood and other modifications," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Toulon, France, 2017, pp. 1910–1919.

[63] A. Hyvärinen and E. Oja, "Independent component analysis: Algorithms and applications," *Neural Netw.*, vol. 13, nos. 4–5, pp. 411–430, Jun. 2000.

[64] A. Van Den Oord, N. Kalchbrenner, and K. Kavukcuoglu, "Pixel recurrent neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1747–1756.

[65] A. Van Den Oord, N. Kalchbrenner, L. Espeholt, K. Kavukcuoglu, O. Vinyals, and A. Graves, "Conditional image generation with PixelCNN decoders," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 4790–4798.

[66] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010.

[67] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[68] L. Zhuang, C.-H. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[69] J. Li, A. Agathos, D. Zaharie, J. M. Bioucas-Dias, A. Plaza, and X. Li, "Minimum volume simplex analysis: A fast algorithm for linear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5067–5082, Sep. 2015.

[70] A. Hyvarinen, "Fast and robust fixed-point algorithms for independent component analysis," *IEEE Trans. Neural Netw.*, vol. 10, no. 3, pp. 626–634, May 1999.

[71] X. Zhang, G. Wen, and W. Dai, "A tensor decomposition-based anomaly detection algorithm for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5801–5820, Oct. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.

[72] A. Krizhevsky, "Learning multiple layers of features from tiny images," Dept. Comput.Sci., Univ. Toronto, Toronto, ON, Canada, Tech. Rep., 2009, pp. 32–33.
[73] K. Tan, Z. Hou, D. Ma, Y. Chen, and Q. Du, "Anomaly detection in hyperspectral imagery based on low-rank representation incorporating a spatial constraint," *Remote Sens.*, vol. 11, no. 13, p. 1578, Jul. 2019.
[74] B. Du and L. Zhang, "A discriminative metric learning based anomaly detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6844–6857, Nov. 2014.
[75] C.-I. Chang, C.-Y. Lin, and P. F. Hu, "Band sampling of hyperspectral anomaly detection in effective anomaly space," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502729.
[76] C.-I. Chang, "Effective anomaly space for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526624.
[77] C.-I. Chang, S. Chen, S. Zhong, and Y. Shi, "Exploration of data scene characterization and 3D ROC evaluation for hyperspectral anomaly detection," *Remote Sens.*, vol. 16, no. 1, p. 135, Dec. 2023.
[78] C.-I. Chang, H. Cao, S. Chen, X. Shang, C. Yu, and M. Song, "Orthogonal subspace projection-based go-decomposition approach to finding low-rank and sparsity matrices for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2403–2429, Mar. 2021.
[79] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.
[80] C.-I. Chang, "Comprehensive analysis of receiver operating characteristic (ROC) curves for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541124.
[81] S. Li, K. Zhang, P. Duan, and X. Kang, "Hyperspectral anomaly detection with kernel isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 319–329, Jan. 2020.



Qingkang He received the B.E. degree in remote sensing science and technology from Central South University, Changsha, China, in 2023. He is currently pursuing the Ph.D. degree in signal and information processing with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include hyperspectral image processing, deep learning, and anomaly detection.



Lianru Gao (Senior Member, IEEE) received the B.S. degree from Tsinghua University, Beijing, China, in 2002, and the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 2007.

He is currently a Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, CAS. He has published more than 300 peer-reviewed articles, and there are more than 190 journal articles included in the Science Citation Index. He obtained 30 national invention patents in China. He was supported by the National Science Foundation as a Distinguished Young Scholar of China in 2023. His research interests include hyperspectral image processing and information extraction.

Dr. Gao is a fellow of the Institution of Engineering and Technology. He received the Outstanding Science and Technology Achievement Prize of CAS in 2016, the Second Prize of the State Scientific and Technological Progress Award in 2018, the IEEE Geoscience and Remote Sensing Society 2025 Highest Impact Paper Award, and the 2021 Outstanding Paper Award from the IEEE Workshop on Hyperspectral Image Processing: Evolution in Remote Sensing. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Michael K. Ng (Senior Member, IEEE) received the B.Sc. and M.Phil. degrees from The University of Hong Kong, Hong Kong, in 1990 and 1992, respectively, and the Ph.D. degree from The Chinese University of Hong Kong, Hong Kong, in 1995.

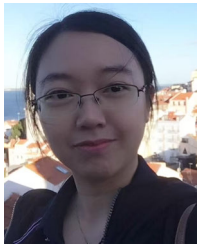
He was a Research Fellow with the Computer Sciences Laboratory, The Australian National University, Canberra, Australia, from 1995 to 1997; and an Assistant/Associate Professor with The University of Hong Kong from 1997 to 2005. He was a Professor/the Chair Professor with the Department of Mathematics, Hong Kong Baptist University, Hong Kong, from 2006 to 2019. He was the Chair Professor with the Research Division of Mathematical and Statistical Science, The University of Hong Kong, from 2019 to 2023. He is currently the Chair Professor of mathematics and data science with Hong Kong Baptist University. His research interests include bioinformatics, image processing, scientific computing, and data mining.

Dr. Ng is selected for the 2017 Class of Fellows of the Society for Industrial and Applied Mathematics. He received the Feng Kang Prize for his significant contributions to scientific computing. He serves on the editorial board member for several international journals.



Yao Liu received the Ph.D. degree in cartography and geographic information systems from the University of Chinese Academy of Sciences, Beijing, China, in 2015.

She is currently a Professor-Level Senior Engineer with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of China, Beijing. Her research interests include remote sensing of inland waters, remote sensing applications on natural resource properties, and data processing methods of Chinese hyperspectral satellites.



Lina Zhuang (Member, IEEE) received the bachelor's degree in geographic information systems and economics from South China Normal University, Guangzhou, China, in 2012, the M.S. degree in cartography and geography information systems from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences (CAS), Beijing, China, in 2015, and the Ph.D. degree in electrical and computer engineering from the Instituto Superior Tecnico, Universidade de Lisboa, Lisbon, Portugal, in 2018.

From 2015 to 2018, she was a Marie Curie Early Stage Researcher of sparse representations and compressed sensing training network (SpaRTaN number 607290) with the Instituto de Telecomunicações. SpaRTaN Initial Training Networks (ITN) is funded under European Union's Seventh Framework Program (FP7-PEOPLE-2013-ITN) call and is part of the Marie Curie Actions–ITN funding scheme. From 2019 to 2021, she was a Research Assistant Professor with the Math Department, Hong Kong Baptist University, Hong Kong. From 2021 to 2022, she was a Research Assistant Professor with the Math Department, The University of Hong Kong, Hong Kong. She is currently a Professor with the Aerospace Information Research Institute, CAS. Her research interests include hyperspectral image restoration, superresolution, and compressive sensing.

Dr. Zhuang currently serves as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and *IET Image Processing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:02:31 UTC from IEEE Xplore. Restrictions apply.